MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONTROLLER, MICROPROCESSOR, KEY CODE RRA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor controller module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.12849 failures per million operating hours for class I or 0.30794 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RRA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/330-1 for class I and M28787/330-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

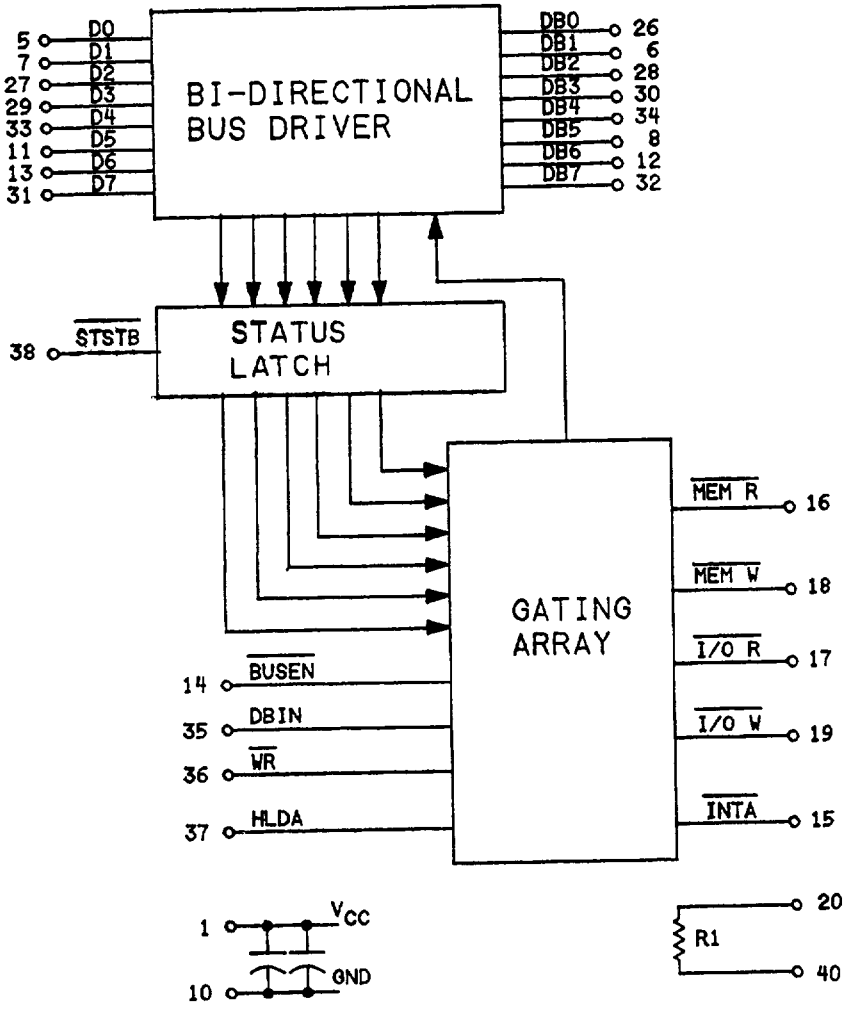
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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UNUSED PINS 2,3,4,9,21, 22,23,24,25,39

FIGURE 1. RRA module function diagram, microprocessor controller.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A11	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	IV	← Must pass all remaining tests →											
C1-C21	Output short circuit current	V	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
D1-D21	Input clamp diode voltage	VI	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
E1-E5	High impedance leakage current, high level	VII	X	-1	100	-1	100	-1	100	-1	100	-1	100	µA
F1-F5	High impedance leakage current, low level	VIII	X	-100	1	-100	1	-100	1	-100	1	-100	1	µA
G1-G13	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G14-G21	Output voltage, high level	IX	X	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	V
H1-H21	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	0	500	mV
J1-J5, J14-J21	Input current, high level	XI	X	0	100	0	100	0	100	0	100	0	100	µA
J6-J13	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
K1	Input current, low level	XII	X	0	-500	0	-500	0	-500	0	-500	0	-500	µA
K2-K15, K17-K19, K21	Input current, low level	XII	X	0	-250	0	-250	0	-250	0	-250	0	-250	µA
K16,K20	Input current, low level	XII	X	0	-750	0	-750	0	-750	0	-750	0	-750	µA
L	Power supply current	XIII	X	21	210	21	210	21	210	21	210	21	210	mA
M1-M118	Functional	XIV	X	← See table XIV →										
N1	Enable delay time (t_{PZH})	XV	S	10	115	10	115	10	115	10	115	10	115	ns
N2	Propagation delay time (t_{PHL})	XV	S	8	80	8	80	8	80	8	80	8	80	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N3	Propagation delay time (t_{PHL} and t_{PLH})	XV	S	7	70	7	70	7	70	7	70	7	70	ns
N4	Enable delay time (t_{PZL}); Propagation delay time (t_{PHL} and t_{PLH})	XV	S	7	70	7	70	7	70	7	70	7	70	ns
N5	Propagation delay time (t_{PHL} and t_{PLH})	XV	S	4	50	4	50	4	50	4	50	4	50	ns
N6	Enable delay time (t_{PZL}); Enable delay time (t_{PZL} and t_{PZH}); Propagation delay time (t_{PLH})	XV	S	4	45	4	45	4	45	4	45	4	45	ns
N7	Propagation delay time (t_{PLH})	XV	S	4	40	4	40	4	40	4	40	4	40	ns
P1-P5	Crosstalk, low level	XVI	Q	0	800	0	800	0	800	0	800	0	800	mV
P6-P21	Crosstalk, low level	XVI	Q	0	750	0	750	0	750	0	750	0	750	mV
Q1-Q21	Crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
R	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
T1-T5	Input capacitance	XX	Q	2	20	2	20	2	20	2	20	2	20	pF
U1-U5	Output capacitance	XXI	Q	2	23	2	23	2	23	2	23	2	23	pF
V1-V16	Input/output capacitance	XXII	Q	3	35	3	35	3	35	3	35	3	35	pF
W1-W3	Setup and hold time	XXIII	S	20	80	20	80	20	80	20	80	20	80	ns
W4-W6	Setup and hold time	XXIII	S	4	45	4	45	4	45	4	45	4	45	ns
W7-W9	Setup and hold time	XXIII	S	4	40	4	40	4	40	4	40	4	40	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W10-W13	Setup and hold time	XXIII	S	7	70	7	70	7	70	7	70	7	70	ns
X1-X32	Setup and hold time	XXIV	S	10	100	10	100	10	100	10	100	10	100	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through S, W, and X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A and C through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1155	1155	mW

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	Class I	Class II	Unit
Input current			
Low level (V_{IL} = 0 V)			
D2,D6	813 max	813 max	µA
$\overline{STSTB}$	542 max	542 max	µA
All others	270 max	270 max	µA
High level (V_{IH} = 5.5 V)			
DB0-DB7	20 max	20 max	µA
All others	100 max	100 max	µA
Output drive current			
Low level (V_{OL} = 0.5 V)			
D0-D7	2	2	mA
All others	10	10	mA
High level (V_{OH} = 2.4 V)			
All others	-1	-1	mA
High level (V_{OH} = 3.3 V)			
D0-D7	-10	-10	mA
Setup and hold times			
t_{SET} (D0-D7 to $\overline{STSTB}$)	14	14	ns
t_{HOLD} (D0-D7 to $\overline{STSTB}$)	7	7	ns
t_{SET} (DB0-DB7 to HLDA)	12	12	ns
t_{HOLD} (DB0-DB7 to HLDA)	22	22	ns

6.5 Input to output relationship. Each microprocessor controller module has the input to output relationship of table II for positive logic.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

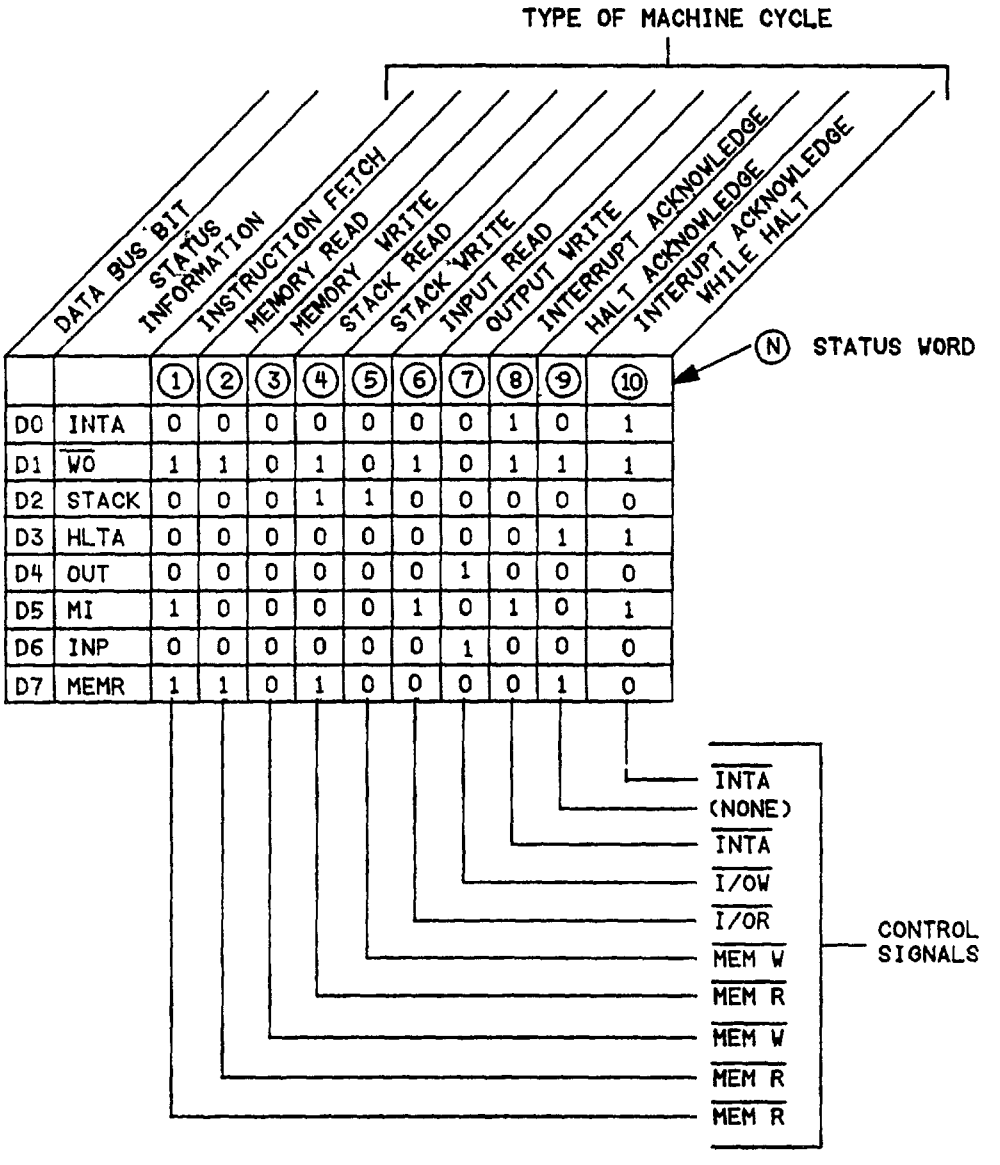
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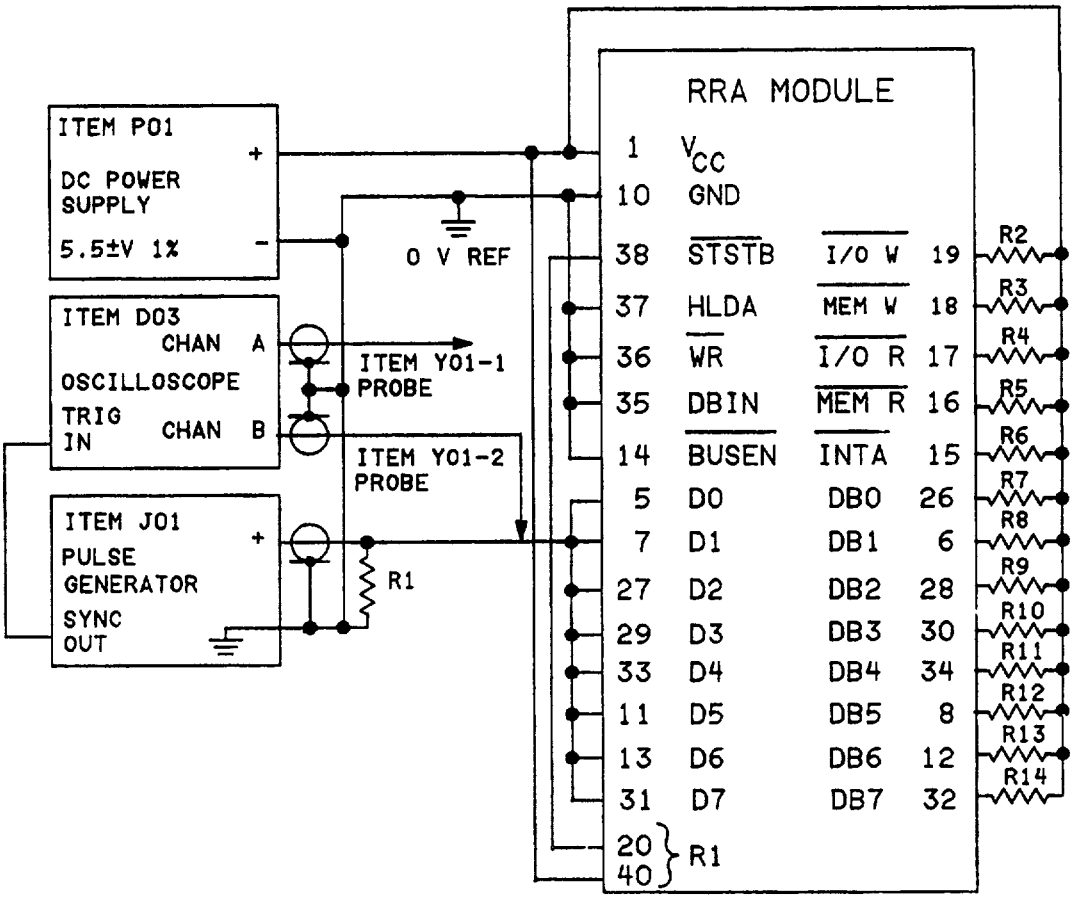
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TABLE II. Input to output relationship (status word chart).



		1	2	3	4	5	6	7	8	9	10
D0	INTA	0	0	0	0	0	0	0	1	0	1
D1	$\overline{WO}$	1	1	0	1	0	1	0	1	1	1
D2	STACK	0	0	0	1	1	0	0	0	0	0
D3	HLTA	0	0	0	0	0	0	0	0	1	1
D4	OUT	0	0	0	0	0	0	1	0	0	0
D5	MI	1	0	0	0	0	1	0	1	0	1
D6	INP	0	0	0	0	0	0	1	0	0	0
D7	MEMR	1	1	0	1	0	0	0	0	1	0

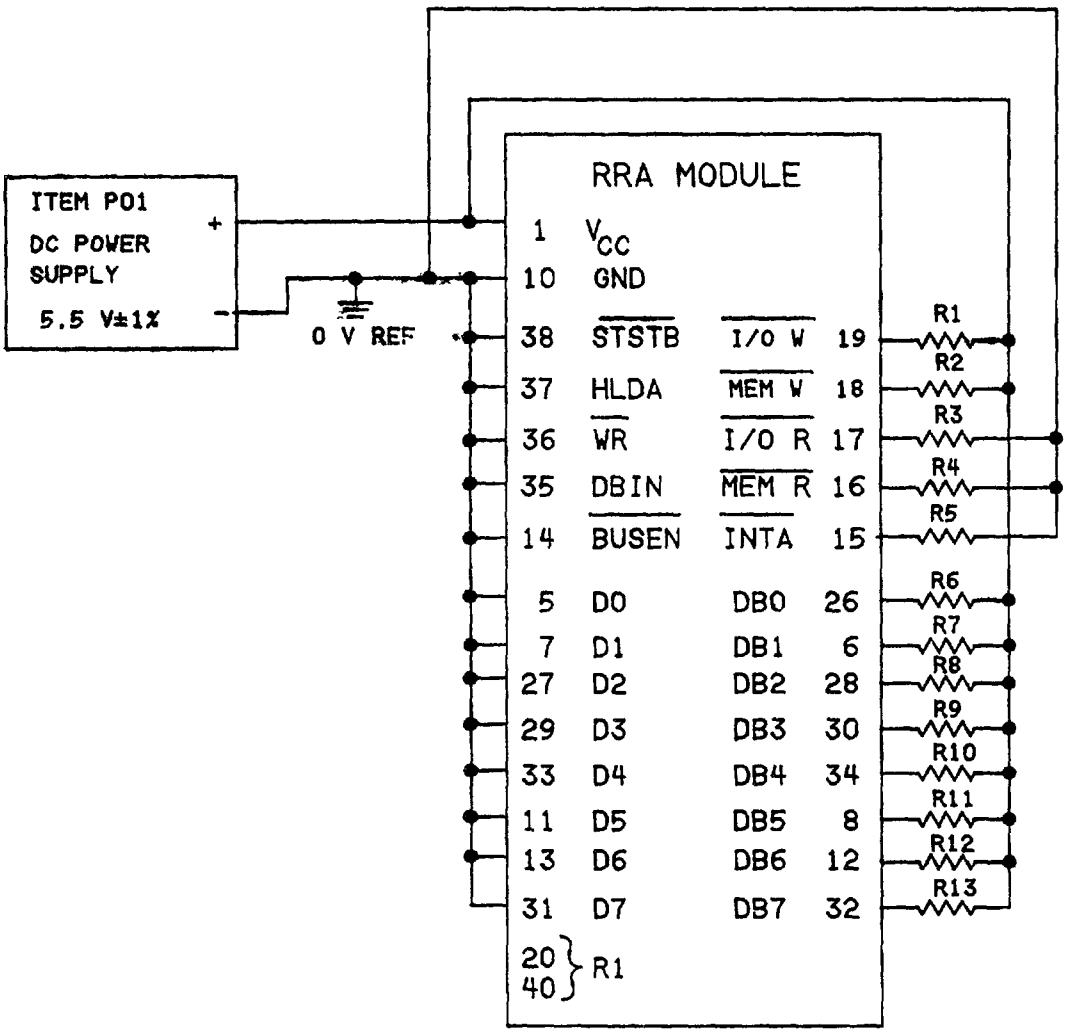
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 - R6 = 550Ω ±1%, 1/4 W.
3. R7 - R14 = 2.75 kΩ ±1%, 1/4 W.
4. Item J01 setup:
   PW = 200 ±20 ns.
   PRP = 400 ±40 ns.
   PA = 3.0 ±0.1 V.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = R6 - R13 = 550Ω ±1%, 1/4 W.
2. R3 - R5 = 2.4 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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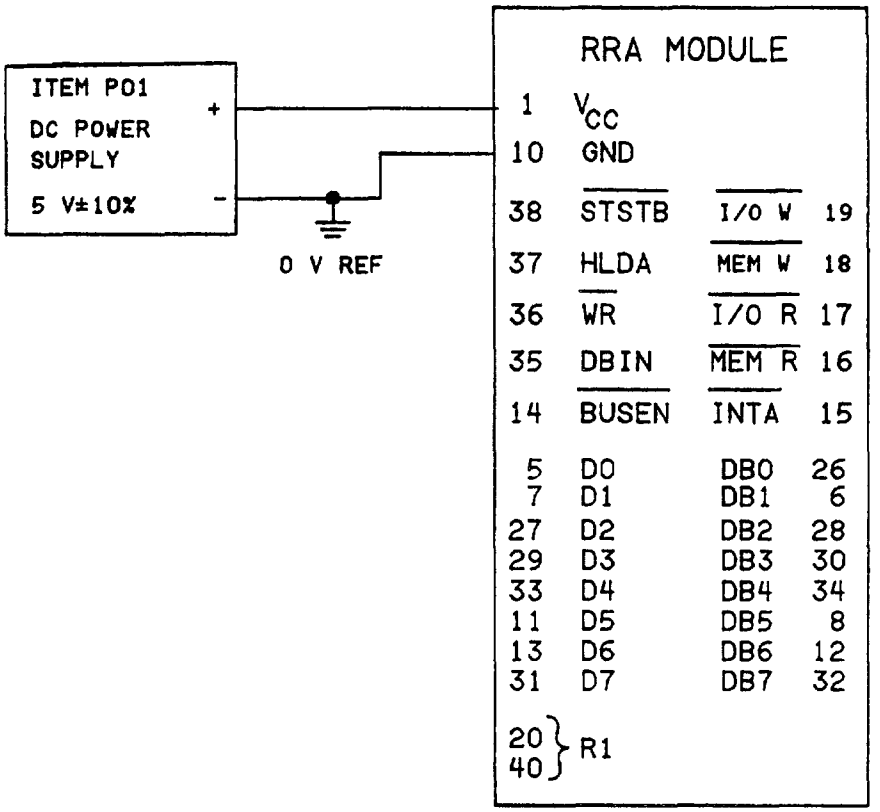


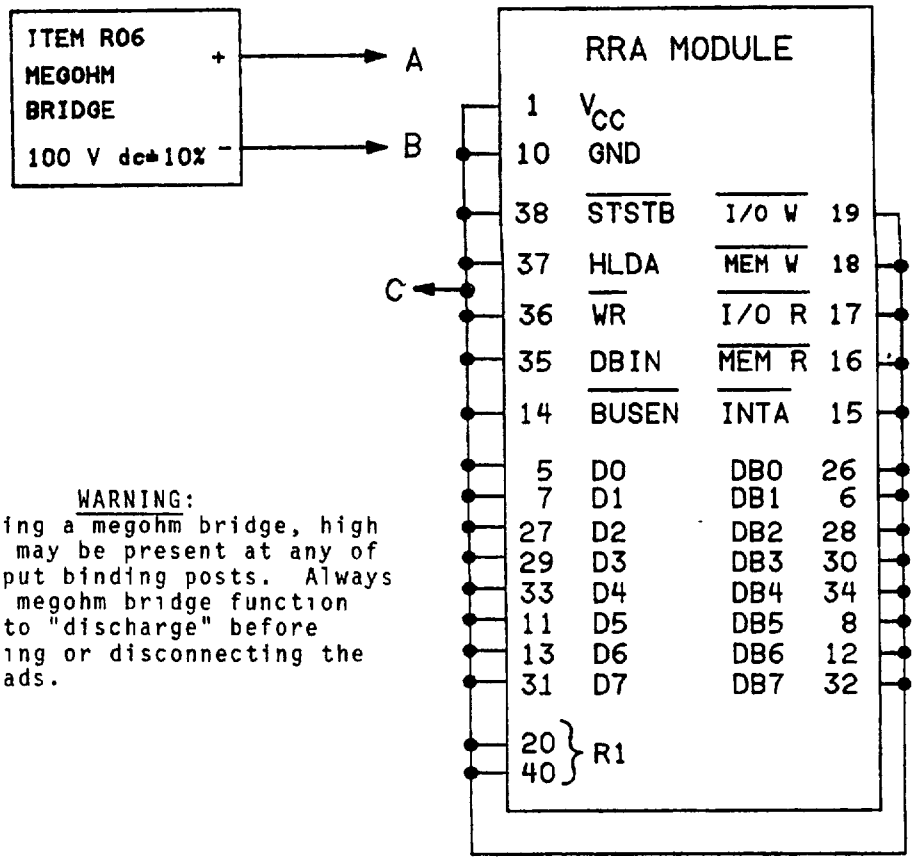
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C,2,3,4,9,21, 22,23,24,25,39	Frame	10	∞	MΩ	5
A2	Isolation, used to unused pins	C	2,3,4,9,21,22,23,24,25,39	"	"	"	"
A3	Isolation, unused to unused pins	2	3,4,9,21,22,23,24,25,39	"	"	"	"
A4	"	3	4,9,21,22,23,24,25,39	"	"	"	"
A5	"	4	9,21,22,23,24,25,39	"	"	"	"
A6	"	9	21,22,23,24,25,39	"	"	"	"
A7	"	21	22,23,24,25,39	"	"	"	"
A8	"	22	23,24,25,39	"	"	"	"
A9	"	23	24,25,39	"	"	"	"
A10	"	24	25,39	"	"	"	"
A11	"	25	39	"	"	"	"

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WARNING:
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

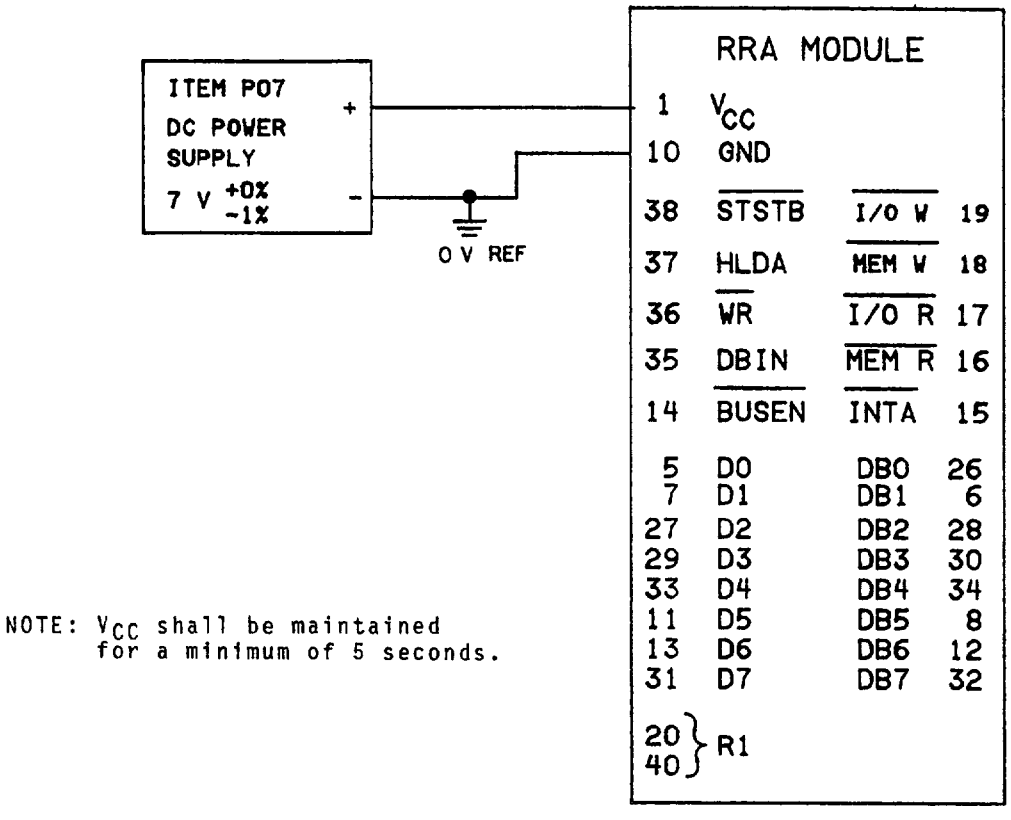
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	←Must pass all remaining tests→			6

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NOTE: V_{CC} shall be maintained for a minimum of 5 seconds.

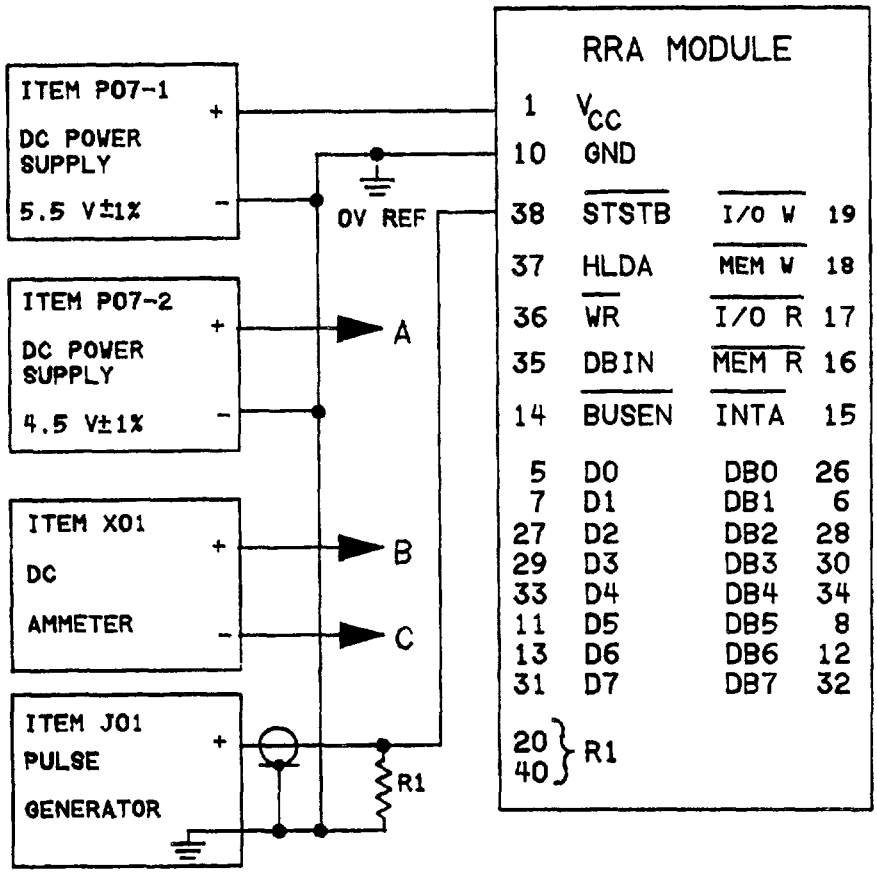
FIGURE 6. V_{CC} stress test circuit for table IV.

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TABLE V. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	$\overline{I/OR}$ Output short circuit current	Open	14,35,36,37,5,7,27,29,33,11,13,31	17	-15	-100	mA	7,8
C2	$\overline{MEMR}$ "	"	"	16	"	"	"	"
C3	$\overline{INTA}$ "	"	"	15	"	"	"	"
C4	$\overline{I/OW}$ "	"	"	19	"	"	"	"
C5	$\overline{MEMW}$ "	5,7,29,11	14,35,36,37,27,33,13,31	18	"	"	"	"
C6	DB0 "	5,7,27,29,33,11,13,31	14,35,36,37	26	"	"	"	"
C7	DB1 "	"	"	6	"	"	"	"
C8	DB2 "	"	"	28	"	"	"	"
C9	DB3 "	"	"	30	"	"	"	"
C10	DB4 "	"	"	34	"	"	"	"
C11	DB5 "	"	"	8	"	"	"	"
C12	DB6 "	"	"	12	"	"	"	"
C13	DB7 "	"	"	32	"	"	"	"
C14	D0 "	14,35,26,6,28,30,34,8,12,32	36,37	5	"	"	"	"
C15	D1 "	"	"	7	"	"	"	"
C16	D2 "	"	"	27	"	"	"	"
C17	D3 "	"	"	29	"	"	"	"
C18	D4 "	"	"	33	"	"	"	"
C19	D5 "	"	"	11	"	"	"	"
C20	D6 "	"	"	13	"	"	"	"
C21	D7 "	"	"	31	"	"	"	"

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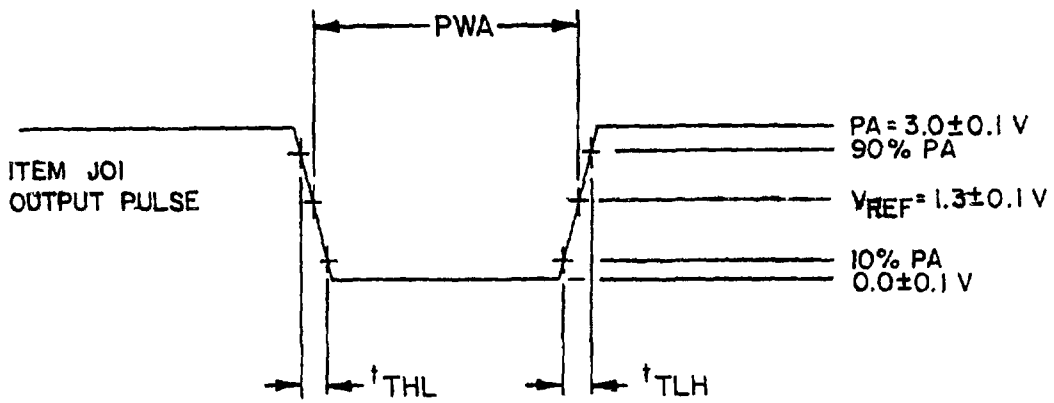


CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. After all connections have been made, clock the module one time for test conditions C1 - C5 and C6 setup.
3. When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

FIGURE 7. Output short circuit current test circuit for table V.

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NOTE: Input pulse parameter:
PWA = 200 ±20 ns
$t_{TLH} = t_{THL}$ = 25 ±10 ns
Single pulse mode.

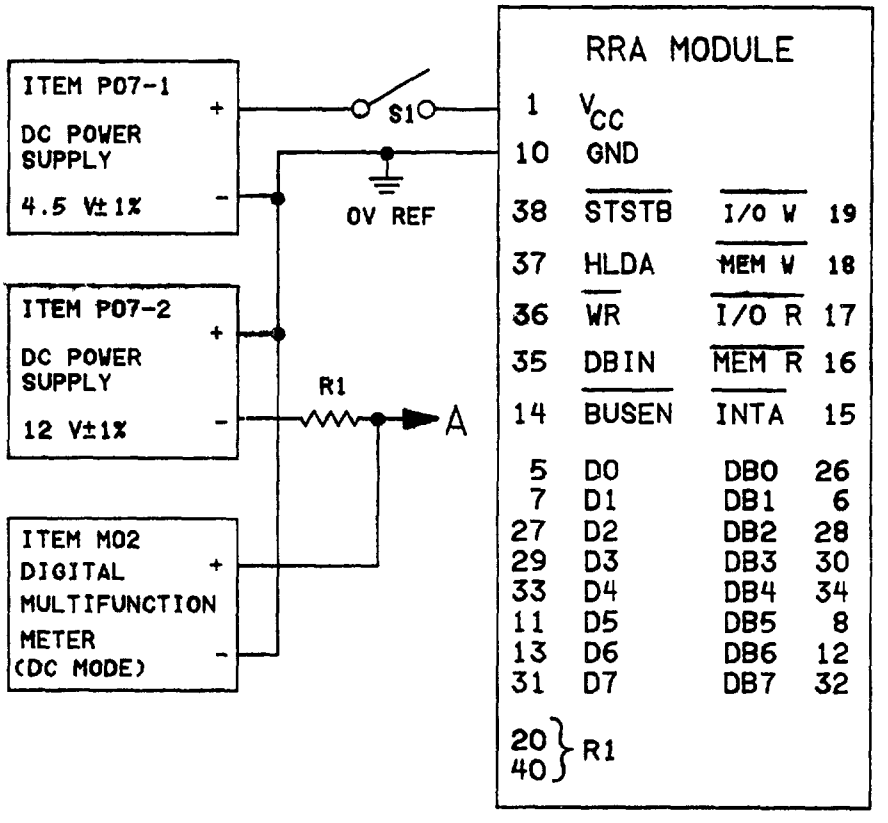
FIGURE 8. Output short circuit current test circuit waveform for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	$\overline{\text{STSTB}}$ Input clamp diode voltage	38	-0.4	-1.2	V	9
D2	HLDA "	37	"	"	"	"
D3	$\overline{\text{WR}}$ "	36	"	"	"	"
D4	DBIN "	35	"	"	"	"
D5	$\overline{\text{BUSEN}}$ "	14	"	"	"	"
D6	DBO "	26	"	"	"	"
D7	DB1 "	6	"	"	"	"
D8	DB2 "	28	"	"	"	"
D9	DB3 "	30	"	"	"	"
D10	DB4 "	34	"	"	"	"
D11	DB5 "	8	"	"	"	"
D12	DB6 "	12	"	"	"	"
D13	DB7 "	32	"	"	"	"
D14	D0 "	5	"	"	"	"
D15	D1 "	7	"	"	"	"
D16	D2 "	27	"	"	"	"
D17	D3 "	29	"	"	"	"
D18	D4 "	33	"	"	"	"
D19	D5 "	11	"	"	"	"
D20	D6 "	13	"	"	"	"
D21	D7 "	31	"	"	"	"

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NOTES
1. R1 = 2.16 kΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. Switch S1 is closed for test conditions D1-D5 and open for test conditions D6-D21.

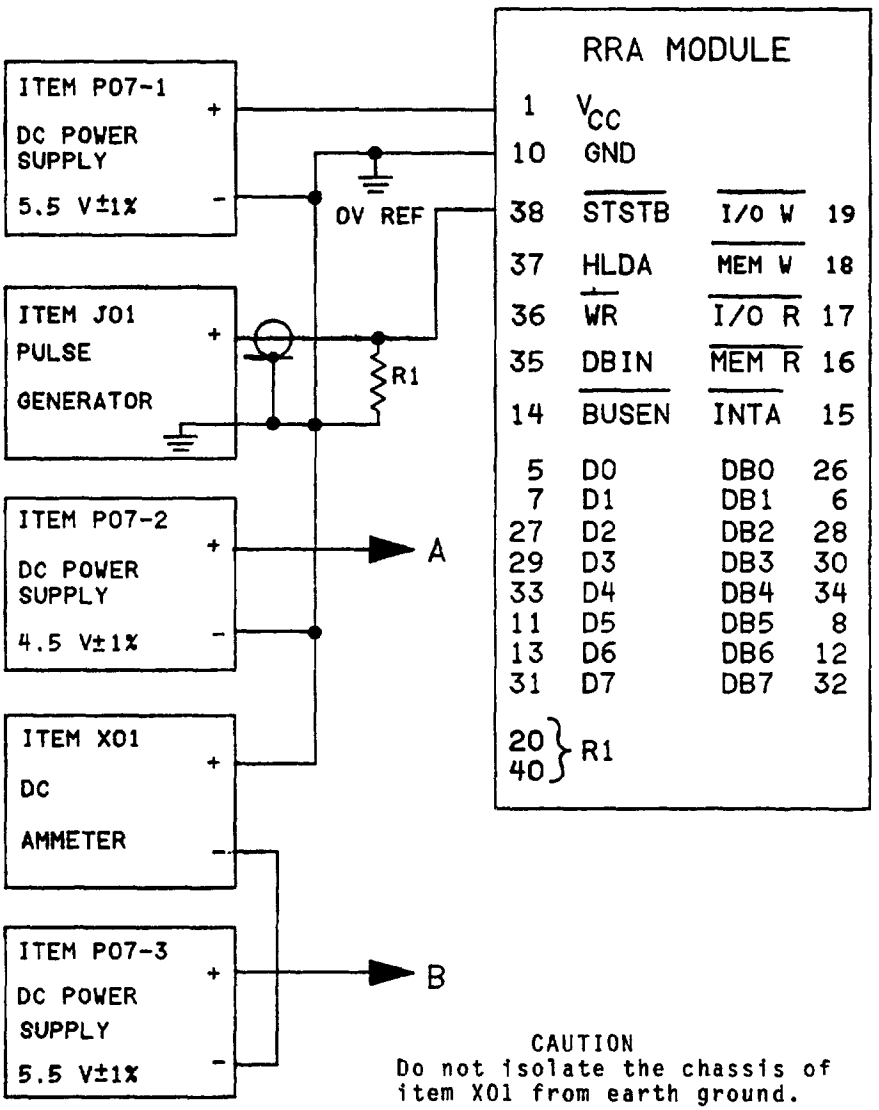
FIGURE 9. Input clamp diode voltage test circuit for table VI.

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TABLE VII. High impedance leakage current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
---	Setup High impedance leakage current, high level	14,36,37,27,13	Open	---	---	---	10
E1	$\overline{\text{INTA}}$	"	15	-1	100	µA	"
E2	$\overline{\text{MEMR}}$	"	"	16	"	"	"
E3	$\overline{\text{I/OR}}$	"	"	17	"	"	"
E4	$\overline{\text{MEMW}}$	"	"	18	"	"	"
E5	$\overline{\text{I/OW}}$	"	"	19	"	"	"

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. For the setup condition, pulse the module one time with a pulse as shown on figure 8.

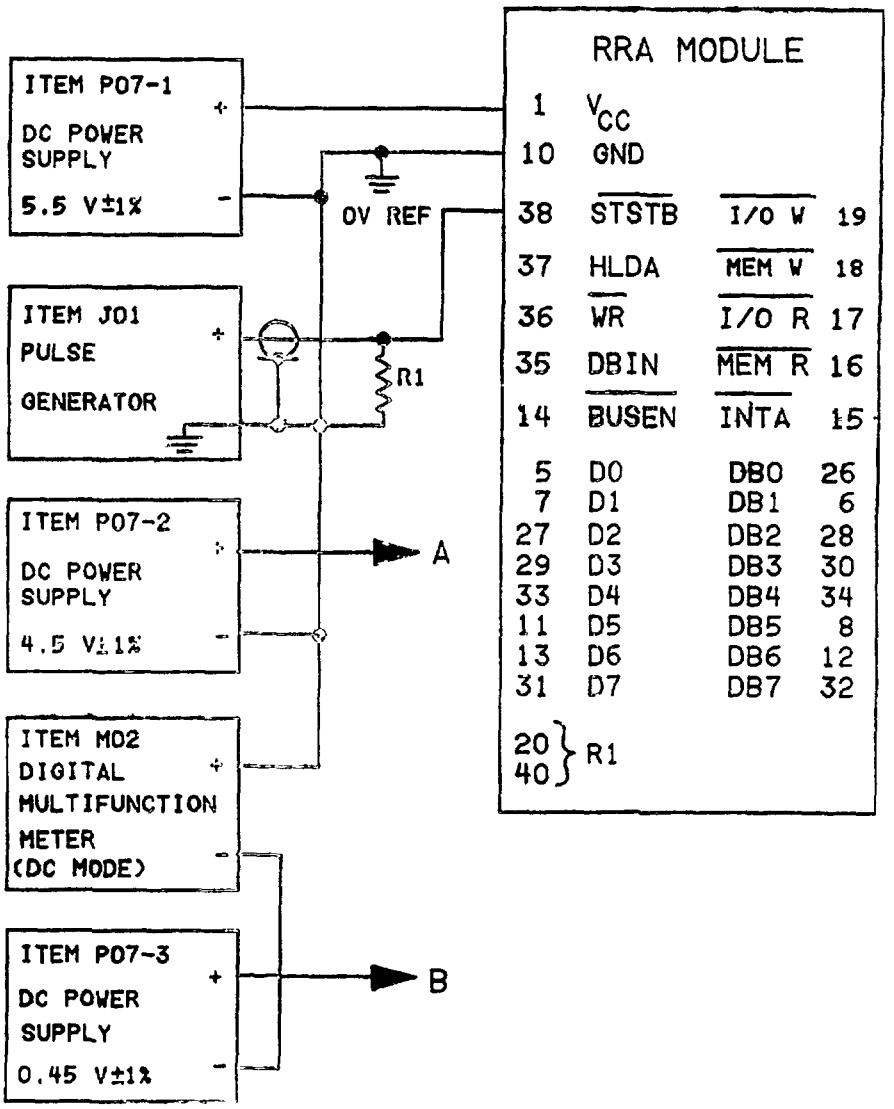
FIGURE 10. High impedance leakage current (high-level) test circuit for table VII.

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TABLE VIII. High impedance leakage current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
---	Setup High impedance leakage current, low level	14,36,37,27,13	Open	---	---	---	11
F1	$\overline{\text{INTA}}$	"	"	15	-100	1	µA	"
F2	$\overline{\text{MEMR}}$	"	"	16	"	"	"	"
F3	$\overline{\text{I/OR}}$	"	"	17	"	"	"	"
F4	$\overline{\text{MEMW}}$	"	"	18	"	"	"	"
F5	$\overline{\text{I/OW}}$	"	"	19	"	"	"	"

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. For the setup condition, pulse the module one time with a pulse as shown on figure 8.

FIGURE 11. High impedance leakage current (low-level) test circuit for table VIII.

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TABLE I. ... output voltage (high-level) tests

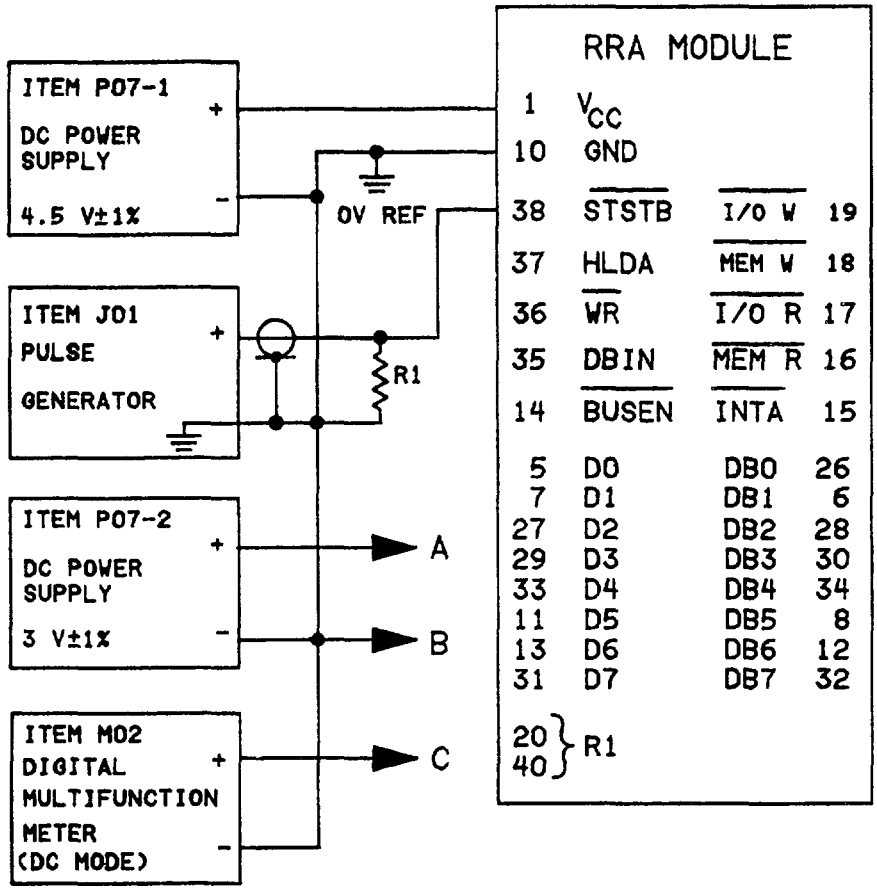
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	$\overline{I/OR}$ Output voltage, high level	36,37,5,7,27,29, 33,11,13,31	14,35	17	2.4	4.5	V	12,13
G2	$\overline{MEMR}$ "	"	"	16	"	"	"	"
G3	$\overline{INTA}$ "	"	"	15	"	"	"	"
G4	$\overline{I/OW}$ "	"	"	19	"	"	"	"
G5	$\overline{MEMW}$ "	"	"	18	"	"	"	"
---	Setup	Open	14,35,5,7,27,29,33, 11,36,37,13,31	---	---	---	---	"
G6	DB0 Output voltage, high level	5	14,35,36,37	26	2.4	4.5	V	"
G7	DB1 "	7	"	6	"	"	"	"
G8	DB2 "	27	"	28	"	"	"	"
G9	DB3 "	29	"	30	"	"	"	"
G10	DB4 "	33	"	34	"	"	"	"
G11	DB5 "	11	"	8	"	"	"	"
G12	DB6 "	13	"	12	"	"	"	"
G13	DB7 "	31	"	32	"	"	"	"
G14	D0 "	14,35,26	37	5	3.3	"	"	"
G15	D1 "	13,35,6	"	7	"	"	"	"
G16	D2 "	14,35,28	"	27	"	"	"	"
G17	D3 "	14,35,30	"	29	"	"	"	"
G18	D4 "	14,35,34	"	33	"	"	"	"
G19	D5 "	14,35,8	"	11	"	"	"	"
G20	D6 "	14,35,12	"	13	"	"	"	"
G21	D7 "	14,35,32	"	31	"	"	"	"

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TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	$\overline{I/OR}$ Output voltage, low level	13,31	14,35,37,27	17	0	500	mV	12,13
H2	$\overline{MEMR}$ "	31	14,35,37,5,27,29,13	16	"	"	"	"
H3	$\overline{INTA}$ "	5,7,11	14,35,37,27,29,13	15	"	"	"	"
H4	$\overline{I/OW}$ "	Open	14,35,36,37,5,7,27	19	"	"	"	"
H5	$\overline{MEMW}$ "	"	14,35,36,37,5,7,27, 29,33,11,13,31	18	"	"	"	"
H6	DB0 "	"	29,33,11,36,37,27, 14,35,7,5,13,31	26	"	"	"	"
H7	DB1 "	"	"	6	"	"	"	"
H8	DB2 "	"	"	28	"	"	"	"
H9	DB3 "	"	"	30	"	"	"	"
H10	DB4 "	"	"	34	"	"	"	"
H11	DB5 "	"	"	8	"	"	"	"
H12	DB6 "	"	"	12	"	"	"	"
H13	DB7 "	"	"	32	"	"	"	"
H14	D0 "	14,35	37,26	5	"	"	"	"
H15	D1 "	"	37,6	7	"	"	"	"
H16	D2 "	"	37,28	27	"	"	"	"
H17	D3 "	"	37,30	29	"	"	"	"
H18	D4 "	"	37,34	33	"	"	"	"
H19	D5 "	"	37,8	11	"	"	"	"
H20	D6 "	"	37,12	13	"	"	"	"
H21	D7 "	"	37,32	31	"	"	"	"

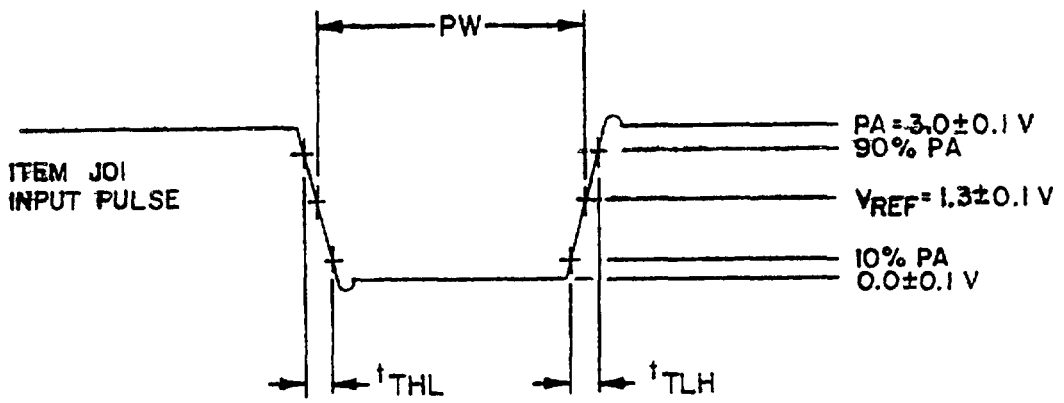
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Pulse the module one time for each test condition G1-G5 and H1-H5 with a pulse as shown on figure 13.
3. Load each output with a load as shown on figure 14.

FIGURE 12. Output voltage (high-level and low-level) test circuit for tables IX and X.

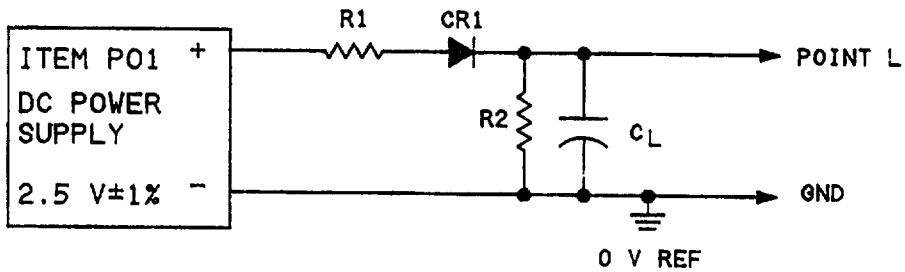
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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 25 ±10 ns.
Single pulse mode.

FIGURE 13. Output voltage (high-level and low-level) test circuit waveform for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select a value for a current flow of 2 mA ±5% out of point L with point L held at 0.5 V for output pins 5, 7, 27, 29, 33, 11, 13, and 31.
   R1 = Select a value for a current flow of 10 mA ±5% out of point L with point L held at 0.5 V for output pins 19, 18, 17, 16, 15, 26, 6, 28, 30, 34, 8, 12, and 32.
3. R2 = 330 kΩ ±1%, 1/8 W for output pins 5, 7, 27, 29, 33, 11, 13, and 31.
   R2 = 2.4 kΩ ±1%, 1/8 W for output pins 19, 18, 17, 16, 15, 26, 6, 28, 30, 34, 8, 12, and 32.
4. C_L = 100 ±10 pF for output pins 5, 7, 27, 29, 33, 11, 13, and 31, except that C_L = 50 ±5 pF for test conditions P1-P21 and Q1-Q21.
   C_L = 150 ±15 pF for output pins 19, 18, 17, 16, 15, 26, 6, 28, 30, 34, 8, 12, and 32, except that C_L = 50 ±5 pF for test conditions P1-P21 and Q1-Q21.

FIGURE 14. RRA logic load circuit.

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TABLE XI. Input current (high-level) test.

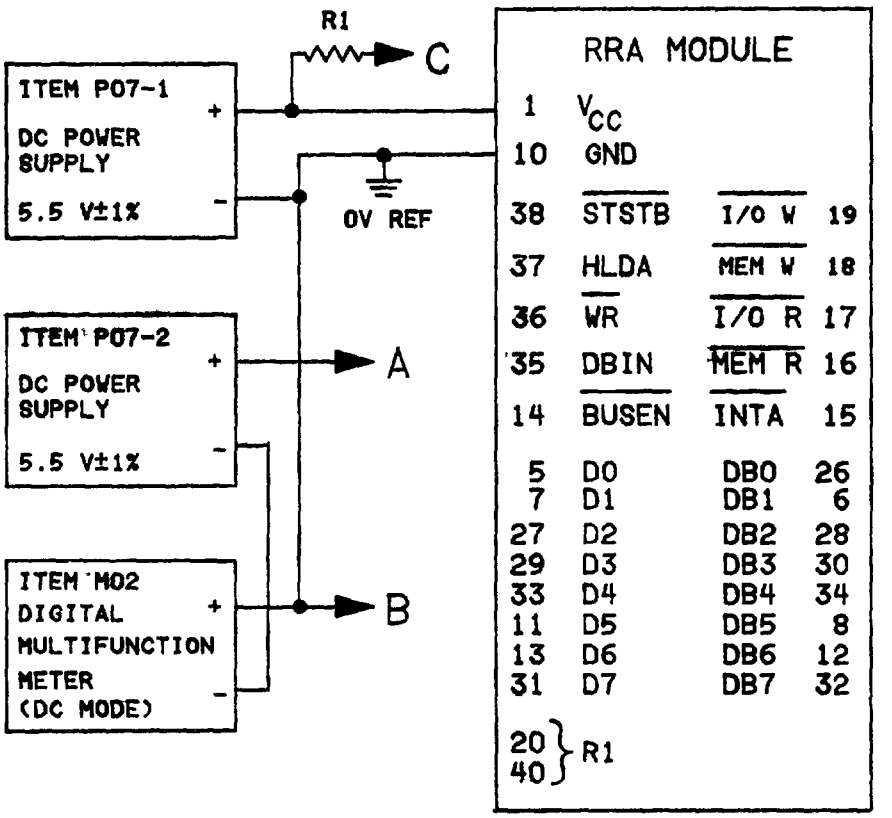
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	$\overline{\text{STSTB}}$ Input current, high level	38	37,36,35,5,7,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	14	0	100	µA	15
J2	HLDA "	37	38,36,35,5,7,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J3	$\overline{\text{WR}}$ "	36	38,37,35,5,7,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J4	DBIN "	35	38,37,36,5,7,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J5	$\overline{\text{BUSEN}}$ "	14	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,30,34,8, 12,32	Open	"	"	"	"
J6	DB0 "	26	38,37,36,35,5,7,27,29,33, 11,13,31,6,28,30,34,8,12,32	14	"	20	"	"
J7	DB1 "	6	38,37,36,35,5,7,27,29,33, 11,13,31,26,28,30,34,8,12, 32	"	"	"	"	"
J8	DB2 "	28	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,30,34,8,12,32	"	"	"	"	"
J9	DB3 "	30	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,34,8,12,32	"	"	"	"	"
J10	DB4 "	34	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,30,8,12,32	"	"	"	"	"
J11	DB5 "	8	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,30,34,12, 32	"	"	"	"	"
J12	DB6 "	12	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,30,34,8,32	"	"	"	"	"
J13	DB7 "	32	38,37,36,35,5,7,27,29,33, 11,13,31,26,6,28,30,34,8,12	"	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J14	D0 Input current, high level	5	38,37,36,35,7,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	14	0	100	µA	15
J15	D1 "	7	38,37,36,35,5,27,29,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J16	D2 "	27	38,37,36,35,5,7,29,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J17	D3 "	29	38,37,36,35,5,7,27,33,11, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J18	D4 "	33	38,37,36,35,5,7,27,29,11, 13,31,26,6,28,30,34,8,12, 32	"	"	"	"	"
J19	D5 "	11	38,37,36,35,5,7,27,29,33, 13,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J20	D6 "	13	38,37,36,35,5,7,27,29,33, 11,31,26,6,28,30,34,8,12,32	"	"	"	"	"
J21	D7 "	31	38,37,36,35,5,7,27,29,33, 11,13,26,6,28,30,34,8,12,32	"	"	"	"	"

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NOTE:
1. R1 = 1 kΩ ±1%, 1/8 W.

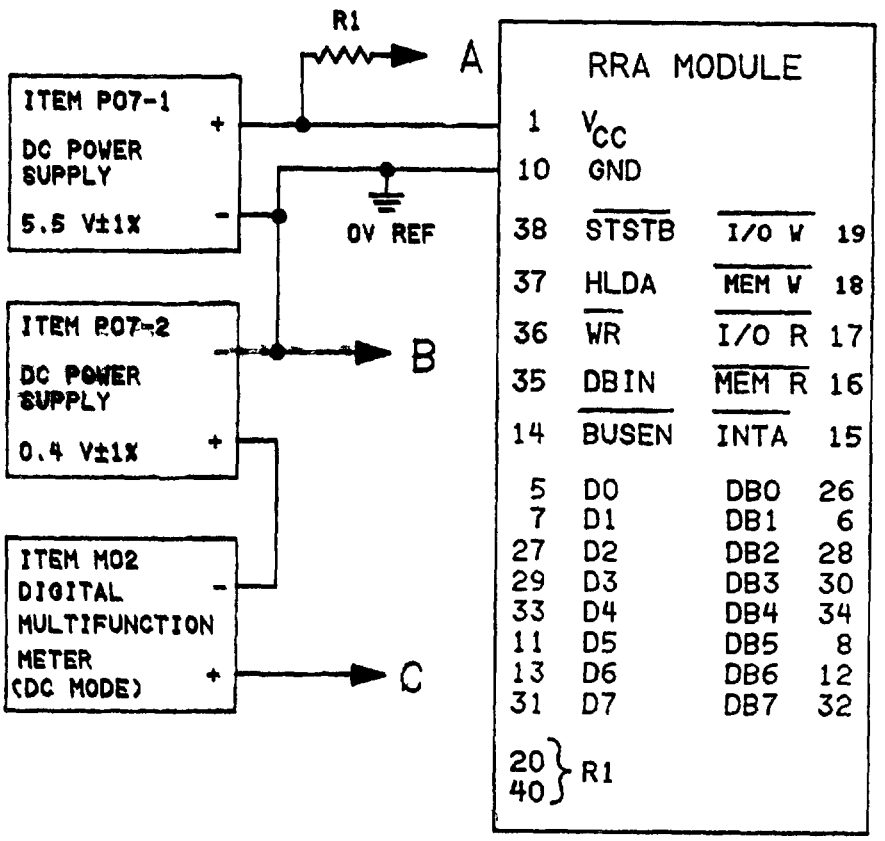
FIGURE 15. Input current (high-level) test circuit for table XI.

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TABLE K... Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	$\overline{\text{STSTB}}$ Input current, low level	Open	Open	38	0	-500	µA	16
K2	HLDA	"	"	37	"	-250	"	"
K3	$\overline{\text{WR}}$	"	"	36	"	"	"	"
K4	DBIN	"	"	35	"	"	"	"
K5	$\overline{\text{BUSEN}}$	"	"	14	"	"	"	"
K6	DB0	14	35	26	"	"	"	"
K7	DB1	"	"	6	"	"	"	"
K8	DB2	"	"	28	"	"	"	"
K9	DB3	"	"	30	"	"	"	"
K10	DB4	"	"	34	"	"	"	"
K11	DB5	"	"	8	"	"	"	"
K12	DB6	"	"	12	"	"	"	"
K13	DB7	"	"	32	"	"	"	"
K14	D0	"	"	5	"	"	"	"
K15	D1	"	"	7	"	"	"	"
K16	D2	"	"	27	"	-750	"	"
K17	D3	"	"	29	"	-250	"	"
K18	D4	"	"	33	"	"	"	"
K19	D5	"	"	11	"	"	"	"
K20	D6	"	"	13	"	-750	"	"
K21	D7	"	"	31	"	-250	"	"

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NOTE: R1 = 1 kΩ ±1%, 1/8 W.

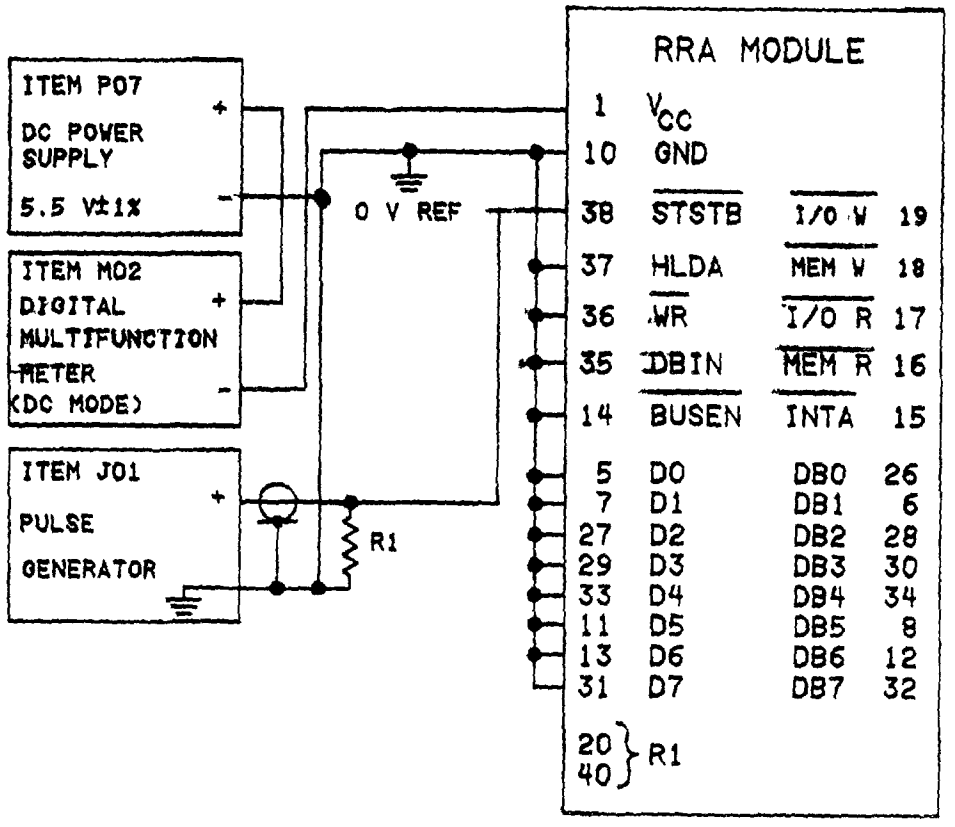
FIGURE 16. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	21	210	mA	17

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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2. Clock the module one time with a pulse as shown on figure 8.

FIGURE 17. Power supply current test circuit for table XIII.

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TABLE XIV. Functional test.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M1	Functional	B	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	18
M2	"	B	A	A	B	B	L	H	L	H	H	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"
M3	"	B	A	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	A	A	B	A	B	B	"
M4	"	B	A	A	B	B	L	H	L	H	H	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"
M5	"	B	A	B	B	B	H	H	H	L	H	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"
M6	"	B	A	A	B	B	L	H	L	H	H	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"
M7	"	A	A	A	B	B	L	H	L	H	H	H	L	L	L	H	L	H	H	A	B	B	B	A	B	A	A	"
M8	"	A	A	A	B	B	L	H	L	H	H	H	L	H	L	H	L	H	H	A	B	A	B	A	B	A	A	"
M9	"	A	A	B	B	B	H	H	H	L	H	H	L	H	L	H	L	H	H	A	B	A	B	A	B	A	A	"
M10	"	A	A	A	B	B	H	H	H	H	H	H	L	H	L	H	L	H	H	A	B	A	B	A	B	A	A	"
M11	"	B	A	A	B	B	L	L	H	H	H	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"
M12	"	B	A	B	B	B	H	H	H	H	L	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"
M13	"	B	A	A	B	B	L	L	H	H	H	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"
M14	"	A	A	A	B	B	L	L	H	H	H	L	L	L	L	L	H	H	H	B	B	B	B	B	A	A	A	"
M15	"	A	A	A	B	B	L	L	H	H	H	H	H	L	H	H	H	H	H	A	A	B	A	A	A	A	A	"
M16	"	A	A	B	B	B	H	H	H	H	L	H	H	L	H	H	H	H	H	A	A	B	A	A	A	A	A	"
M17	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	A	A	B	A	B	A	"
M18	"	B	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	A	A	B	A	B	A	"
M19	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	A	A	B	A	B	A	"
M20	"	A	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	A	A	B	A	B	A	"
M21	"	A	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	B	B	B	B	A	A	"

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M22	Functional	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	B	B	B	B	A	A	18
M23	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	B	B	B	B	B	A	A	"
M24	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H	X	"
M25	"	A	B	B	A	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M26	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M27	"	B	B	B	B	B	H	H	H	L	H	X	X	X	X	X	X	X	X	B	B	B	B	A	B	A	A	"
M28	"	B	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	A	B	B	B	A	A	"
M29	"	B	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	B	A	B	A	A	A	A	"
M30	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	B	A	B	A	A	A	A	"
M31	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M32	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M33	"	A	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	H	X	X	X	H	X	"
M34	"	B	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	A	B	B	B	A	A	"
M35	"	B	B	B	B	B	H	H	H	H	L	X	X	X	X	X	X	X	X	B	B	B	B	B	B	A	A	"
M36	"	B	B	B	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	B	A	B	B	A	A	"
M37	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	B	A	B	B	A	A	"
M38	"	B	B	A	B	B	H	L	H	H	H	X	X	X	X	X	X	X	X	B	A	B	B	B	B	A	A	"
M39	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	B	B	B	B	A	B	"
M40	"	B	B	A	B	B	H	H	L	H	H	X	X	X	X	X	X	X	X	A	B	A	B	B	B	A	B	"
M41	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	B	A	B	B	B	A	B	"
M42	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	A	B	B	B	A	B	"

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M43	Functional	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	A	A	B	B	B	A	B	18
M44	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	A	A	B	B	A	A	B	"
M45	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	A	A	A	B	B	B	A	B	"
M46	"	B	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	B	A	B	B	B	B	A	B	"
M47	"	B	B	A	B	B	L	H	H	H	H	X	X	X	X	X	X	X	X	B	A	B	B	B	B	B	B	"
M48	"	B	B	B	B	B	H	H	H	L	H	X	X	X	X	X	X	X	X	B	A	B	B	A	A	A	A	"
M49	"	A	B	B	B	B	H	H	H	L	H	X	X	X	X	X	X	X	X	X	A	X	X	A	A	A	A	"
M50	"	A	B	B	B	B	H	H	H	L	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M51	"	A	B	B	B	A	X	X	X	X	X	B	A	B	A	B	A	B	A	X	X	H	X	X	X	H	X	"
M52	"	A	B	B	B	A	X	X	X	X	X	A	B	A	B	A	B	A	B	X	X	H	X	X	X	H	X	"
M53	"	A	B	A	B	A	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"
M54	"	A	B	A	A	B	H	L	H	H	H	B	B	B	B	B	B	B	B	L	L	L	L	L	L	L	L	"
M55	"	A	B	A	A	B	H	L	H	H	H	B	B	A	A	A	A	B	B	L	L	H	H	H	H	L	L	"
M56	"	A	B	A	A	B	H	L	H	H	H	B	A	A	B	B	B	B	B	L	H	H	L	L	L	L	L	"
M57	"	A	B	A	A	B	H	L	H	H	H	B	A	A	A	B	B	B	B	L	H	H	H	L	L	L	L	"
M58	"	A	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M59	"	B	B	A	B	B	H	L	L	H	H	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"
M60	"	B	B	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"
M61	"	B	B	A	B	B	H	H	L	H	H	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"
M62	"	A	B	A	B	B	H	H	L	H	H	H	L	H	H	L	H	L	H	A	B	A	A	B	A	B	A	"
M63	"	A	B	A	B	B	H	H	L	H	H	H	H	H	H	H	H	H	H	A	A	A	A	A	A	A	A	"

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M64	Functional	A	B	B	B	B	H	H	H	H	L	H	H	H	H	H	H	H	H	A	A	A	A	A	A	A	A	18
M65	"	A	B	B	B	B	H	H	H	H	L	L	L	L	L	L	L	L	L	B	B	B	B	B	B	B	B	"
M66	"	A	A	B	B	B	H	H	H	H	L	L	H	L	L	L	H	H	H	B	A	B	B	B	A	A	A	"
M67	"	A	A	B	B	B	H	H	H	H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M68	"	A	A	B	A	B	H	H	H	H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M69	"	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M70	"	A	B	B	B	B	H	H	H	H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M71	"	B	B	A	B	B	L	H	L	H	H	X	X	X	X	X	X	X	X	A	B	B	B	B	A	A	A	"
M72	"	A	B	A	B	B	L	H	L	H	H	H	L	L	L	L	H	H	H	A	B	B	B	B	A	A	A	"
M73	"	A	B	A	B	B	L	H	L	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M74	"	A	B	A	A	B	L	H	L	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M75	"	A	A	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	H	X	X	X	H	X	"
M76	"	A	A	A	A	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M77	"	A	B	A	B	B	H	H	H	H	H	X	X	X	X	X	X	X	X	X	X	H	X	X	X	H	X	"
M78	"	B	B	A	B	B	H	L	H	H	H	X	X	X	X	X	X	X	X	B	B	A	B	A	A	B	A	"
M79	"	A	B	A	B	B	H	L	H	H	H	L	L	H	L	H	H	L	H	B	B	A	B	A	A	B	A	"
M80	"	A	B	A	B	B	H	L	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M81	"	A	B	A	A	B	H	L	H	H	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M82	"	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M83	"	A	A	A	B	A	X	X	X	X	X	X	B	B	B	X	B	X	X	X	X	X	X	X	X	X	X	"
M84	"	A	A	A	A	A	X	X	X	X	X	X	B	B	B	X	B	X	X	X	X	X	X	X	X	X	X	"

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TABLE XIV. Functional test - Continued.

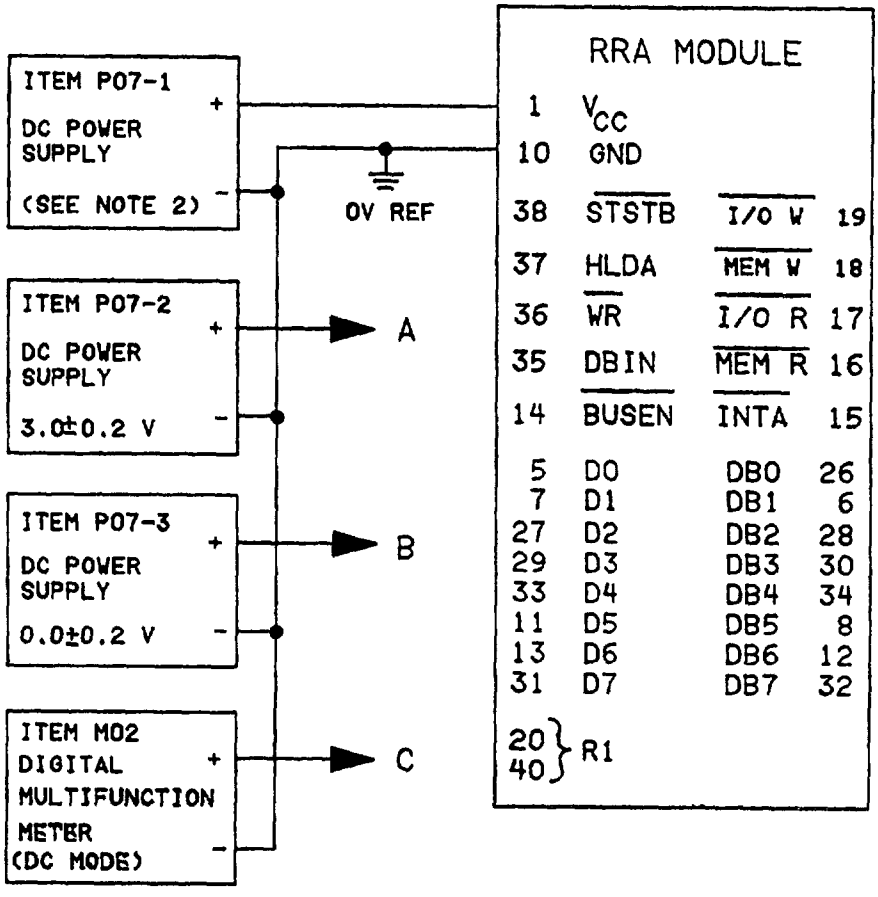
Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M85	Functional	A	A	A	B	A	X	X	X	X	X	X	B	B	B	X	B	X	X	X	X	X	X	X	X	X	X	18
M86	"	A	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"
M87	"	B	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"
M88	"	A	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	A	A	A	B	A	"
M89	"	A	X	X	B	B	X	X	X	X	X	X	H	X	X	X	H	H	H	X	A	X	X	X	A	A	A	"
M90	"	A	X	X	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	X	A	A	A	"
M91	"	A	X	X	B	B	X	X	X	X	X	H	H	H	H	H	H	H	H	A	A	A	A	A	A	A	A	"
M92	"	A	X	X	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	B	B	B	"
M93	"	A	X	X	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	B	B	B	"
M94	"	A	X	X	B	B	X	X	X	X	X	L	X	L	L	L	X	X	L	B	B	B	B	B	B	B	B	"
M95	"	A	X	X	B	B	X	X	X	X	X	L	L	L	L	L	L	L	L	B	B	B	B	B	B	B	B	"
M96	"	A	X	X	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	B	B	B	"
M97	"	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	A	X	X	B	X	"
M98	"	A	B	B	B	B	X	X	X	X	X	X	X	X	H	X	X	X	X	X	B	B	A	X	X	B	X	"
M99	"	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	X	B	X	B	B	"
M100	"	A	B	B	B	B	X	X	X	X	X	X	X	X	X	L	X	X	L	X	B	B	X	B	X	B	B	"
M101	"	A	B	B	A	A	X	X	X	X	X	A	X	X	X	X	X	A	A	H	X	X	X	X	X	H	H	"
M102	"	A	B	B	A	A	X	X	X	X	X	B	X	X	X	X	X	B	B	L	X	X	X	X	X	L	L	"
M103	"	A	B	B	A	A	X	X	X	X	X	A	A	A	X	X	X	A	A	H	H	H	X	X	X	H	H	"
M104	"	A	B	B	A	A	X	X	X	X	X	B	B	B	X	X	X	B	B	L	L	L	X	X	X	L	L	"
M105	"	B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	B	X	X	A	X	X	"

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Test figure
		Pins					Pins					System bus pins								Processor bus pins								
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31	
M106	Functional	A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	B	X	X	A	X	X	18
M107	"	A	B	A	A	B	X	X	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M108	"	A	A	A	A	B	X	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M109	"	B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	B	B	X	A	X	A	"
M110	"	A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	B	B	X	A	X	A	"
M111	"	A	B	A	A	B	X	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M112	"	A	A	A	A	B	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M113	"	B	B	A	B	B	X	H	X	X	X	X	X	X	X	X	X	X	X	X	B	X	X	X	A	X		"
M114	"	A	B	A	B	B	X	H	X	X	X	X	X	X	X	X	X	X	X	X	B	X	X	X	A	X		"
M115	"	A	B	A	A	B	L	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M116	"	A	A	A	A	B	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
M117	"	B	A	A	B	B	X	X	X	X	X	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	"
M118	"	B	A	A	A	B	X	X	X	X	X	A	A	A	A	A	A	A	A	H	H	H	H	H	H	H	H	"

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NOTES
1. Connect point C to the module pins specified in table XIV.
2. Adjust item P07-1 as follows:
   V_{CC} = 4.5 V ±1% for test condition M1-M118 (test conditions are performed twice).
   V_{CC} = 5.5 V ±1% for test condition M1-M118 (test conditions are performed twice).
3. Output levels:
   Processor bus (D0-D7):
   H = 3.3 V minimum.
   L = 0.5 V maximum.
   System bus, status lines (DB0-DB7, $\overline{INTA}$, $\overline{MEMR}$, $\overline{MEMW}$, $\overline{I/OW}$, $\overline{I/OR}$).
   H = 2.4 V minimum
   L = 0.5 V maximum.
4. Cycle times to be 200 ±20 ns. Output levels are to be verified at 176 ns referenced to the start of the cycle time.

FIGURE 18. Functional test circuit for table XIV.

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N1	Enable delay time (t_{PZH}) DBIN to D0-D7	B	A	A	B	X	X	X	X	X	X	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	10	115	ns	19
		B	A	A	A	B	X	X	X	X	X	A	A	A	A	A	A	A	A	H	H	X	H	H	H	X	H	10	115	"	"
N2	Propagation delay time (t_{PHL}), $\overline{STSTB}$ to $\overline{MEMR}$, $\overline{I/OR}$ and $\overline{INTA}$	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	8	80	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	B	A	B	A	A	"	"	"	"
		A	A	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	B	A	B	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	B	A	B	A	A	"	"	"	"
		B	A	A	B	B	L	L	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
		B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	A	B	B	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	B	A	B	A	B	A	B	A	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	A	B	A	B	A	B	A	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	A	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	A	A	A	A	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	A	A	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	B	A	A	A	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		B	B	A	B	B	X	X	L	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"	"	"	"
N3	Propagation delay time (t_{PHL} and t_{PLH}), $\overline{WR}$ to $\overline{I/OW}$ and $\overline{MEMW}$	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	7	70	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	A	A	B	A	B	B	"	"	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"	"	"	"

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N3	Propagation delay time (t_{PHL} and t_{PLH}), $\overline{WR}$) to I/OW and MEMW	B	A	B	B	B	X	X	X	L	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	7	70	ns	19
		B	A	A	B	B	X	X	X	H	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	B	A	B	A	A	"	"	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X									"	"	"	"
		B	A	B	B	B	X	X	X	X	L	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
		B	A	A	B	B	X	X	X	X	H	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
N4	Enable delay time (t_{PZL}), DBIN to DO-D7, Propagation delay time (t_{PHL} and t_{PLH}), DBO-DB7 to DO-D7	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	A	B	B	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	B	A	B	A	B	A	B	A	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	A	B	A	B	A	B	A	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	A	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	B	B	B	B	B	B	L	L	L	L	L	L	L	L	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	A	A	A	A	B	B	X	X	H	H	H	H	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	A	A	B	B	B	B	B	X	H	X	L	L	L	X	X	"	"	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	H	X	X	X	H	X	X	H	A	B	B	B	A	B	A	A	"	"	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	H	X	X	B	B	B	B	B	A	A	A	"	"	"	"

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N4	Enable delay time (t_{PZL}), DBIN to DO-D7,	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	A	X	X	B	X	7	70	ns	19
	Propagation delay time (t_{PHL} and t_{PLH}), DB0-DB7 to DO-D7	A	B	B	B	B	X	X	X	X	X	X	X	X	H	X	X	X	X	X	B	B	A	X	X	B	X	"	"	"	"
		B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	X	B	X	B	B	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	L	X	B	B	X	X	X	B	B	"	"	"	"
		A	B	B	A	A	X	X	X	X	X	A	X	X	X	X	X	A	A	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	A	A	X	X	X	X	X	B	X	X	X	X	X	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	A	A	X	X	X	X	X	A	A	A	X	X	X	A	A	H	X	X	X	X	X	H	H	"	"	"	"
		A	B	B	A	A	X	X	X	X	X	B	B	B	X	X	X	B	B	L	L	L	X	X	X	L	L	"	"	"	"
N5	Propagation delay time (t_{PHL} and t_{PLH}), DO-D7 to DB0-DB7	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	4	50	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	H	X	X	X	X	X	A	B	A	A	A	B	A	A	"	"	"	"
		B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	B	B	B	A	A	A	"	"	"	"
		A	A	A	B	B	X	X	X	X	X	H	H	X	H	H	X	X	X	A	A	B	A	A	A	A	A	"	"	"	"
		B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"	"	"	"
		B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	A	A	A	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	A	A	A	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	L	L	L	L	L	L	L	L	B	B	B	B	B	B	B	B	"	"	"	"
		A	A	B	B	B	X	X	X	X	X	X	H	X	X	X	H	H	H	B	A	B	B	B	A	A	A	"	"	"	"

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N6	Enable delay time (t_{PZL}), STSTB to DB0-DB7;	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	A	B	A	A	4	45	ns	19
	Enable delay time (t_{PZL} and t_{PZH}), BUSEN to DB0-DB7;	A	A	A	B	B	X	X	X	X	X	X	L	L	L	X	L	X	X	A	B	B	B	A	B	A	A	"	"	"	"
	Propagation delay time (t_{PLH}), DBIN to $\overline{IOR}$, $\overline{MEMR}$ and $\overline{INTA}$	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	A	B	B	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	A	A	A	A	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	B	A	B	A	B	A	B	A	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	B	B	A	X	X	X	X	X	A	B	A	B	A	B	A	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	A	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	B	A	A	A	A	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	A	A	B	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	B	A	A	A	B	B	B	B	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	B	B	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	H	H	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"	"	"	"
		B	H	H	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"	"	"	"
		A	H	H	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	A	A	B	A	B	A	"	"	"	"
		A	H	H	B	B	X	X	X	X	X	H	L	H	H	L	H	L	H	A	B	A	A	B	A	B	A	"	"	"	"
		B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	B	A	A	A	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	B	B	A	A	A	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	B	A	A	B	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

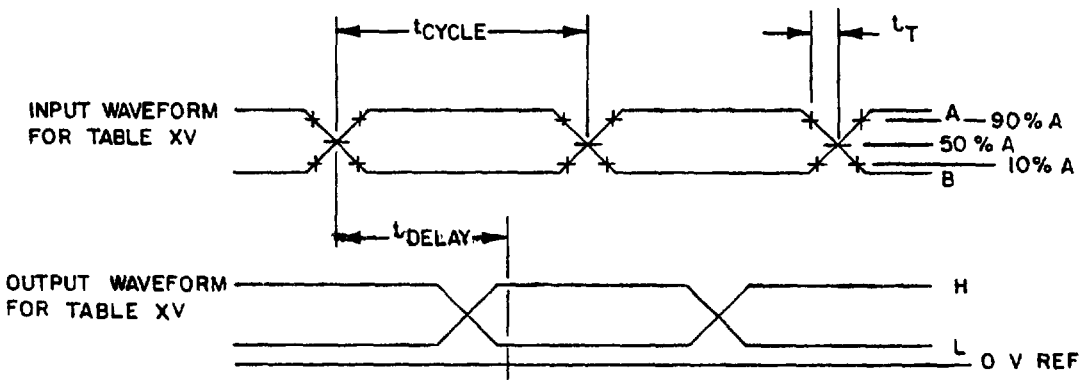
Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N6	Enable delay time (t_{PZL}), STSTB to DB0-DB7;	A	A	A	B	B	X	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	4	45	ns	19
	Enable delay time (t_{PZL} and t_{PZH}), BUSEN to DB0-DB7;	A	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"	"	"	"
		B	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	B	B	A	A	A	B	A	"	"	"	"
	Propagation delay time (t_{PLH}), DBIN to $\overline{I/OR}$, $\overline{MEMR}$ and $\overline{INTA}$	A	X	X	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	A	A	A	A	A	A	A	A	"	"	"	"
		A	X	X	B	B	X	X	X	X	X	X	H	X	X	X	X	H	X	X	A	X	X	X	A	A	A	"	"	"	"
		A	X	X	B	A	X	X	X	X	X	X	H	X	X	X	X	X	X	X	A	X	X	X	A	A	A	"	"	"	"
		A	X	X	B	B	X	X	X	X	X	X	H	X	X	X	X	X	X	A	A	A	A	A	A	A	A	"	"	"	"
		A	X	X	B	A	X	X	X	X	X	X	H	X	X	X	X	X	X	B	B	B	B	B	B	B	B	"	"	"	"
		A	X	X	B	A	X	X	X	X	X	X	H	X	X	X	X	X	X	B	B	B	B	B	B	B	B	"	"	"	"
		A	X	X	B	B	X	X	X	X	X	L	X	L	L	X	L	X	L	B	B	B	B	B	B	B	B	"	"	"	"
		B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	B	X	B	X	B	B	"	"	"	"
		A	B	B	B	B	X	X	X	X	X	X	X	X	X	L	X	X	X	X	B	B	X	B	X	B	B	"	"	"	"
N7	Propagation delay time (t_{PLH}), HLDA to $\overline{I/OR}$, $\overline{INTA}$ and $\overline{MEMR}$	B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	B	X	X	A	X	X	"	40	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	B	X	X	A	X	X	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	A	A	A	B	X	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	B	B	X	A	X	A	"	"	"	"
		A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	B	B	X	A	X	A	"	"	"	"
		A	B	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	A	A	A	B	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	X	X	B	X	"	"	"	"

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TABLE XV. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Inputs					Outputs					I/O Ports																Limits		Unit	Test figure
		Pins					Pins					System bus pins								Processor bus pins								Min	Max		
		38	37	36	35	14	17	16	15	19	18	26	6	28	30	34	8	12	32	5	7	27	29	33	11	13	31				
N7	Propagation delay time (t_{PLH}), HLDA to $\overline{I/OR}$, $\overline{INTA}$ and $\overline{MEMR}$	A	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	X	X	X	B	X	4	40	ns	19
		A	B	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"
		A	A	A	A	B	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"	"	"	"

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NOTES

1. t_{CYCLE} = 200 ±20 ns.
2. t_T = 10 ±2 ns.
3. A = 3.0 ±0.2 V.
4. B = 0.0 ±0.2 V.
5. t_{DELAY} = 115 ±1 ns for test condition N1.
   80 ±1 ns for test condition N2.
   70 ±1 ns for test conditions N3 and N4.
   50 ±1 ns for test condition N5.
   45 ±1 ns for test condition N6.
   40 ±1 ns for test condition N7.
6. H = 3.3 V minimum for processor bus (D0-D7).
   = 2.4 V minimum for system bus (DB0-DB7) and status lines ($\overline{INTA}$, $\overline{MEMR}$, $\overline{MEMW}$, $\overline{I/OR}$, and $\overline{I/OW}$).
7. L = 0.5 V maximum.
8. Enable and propagation delay times are verified by comparing the output states (H, L) to those listed in table XV for test conditions N1-N7 at the times specified by t_{DELAY}

FIGURE 19. Enable delay time (t_{PZH} and t_{PZL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms and timing parameters for table XV.

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TABLE XVI. Crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	13,31	14,35,37,27,7	Open	Open	---	---	---	20,21
P1	$\overline{I/OR}$ Crosstalk, low level	Open	14,35,37	5,7,29,33,11,13,31,27	17	0	800	mV	20,22
P2	$\overline{INTA}$ Crosstalk, low level	"	14,35,37	5,7,29,33,11,13,31,27	15	0	800	mV	20,22
---	Setup	"	14,35,37,36,5,7,27	Open	Open	---	---	---	20,21
P3	$\overline{I/OW}$ Crosstalk, low level	"	14,35,36,37	5,7,27,29,33,11,13,31	19	0	800	mV	20,22
---	Setup	"	14,35,36,37,5,7,27,29,33,11,13,31	Open	Open	---	---	---	20,21
P4	$\overline{MEMW}$ Crosstalk, low level	"	14,35,36,37	5,7,27,29,33,11,13,31	18	0	800	mV	20,22
---	Setup	31	14,35,37,5,27,29,13,7	Open	---	---	---	---	20,21
P5	$\overline{MEMR}$ Crosstalk, low level	Open	14,35,37	5,7,27,29,33,11,13,31	16	0	800	mV	20,22
P6	DB0 "	"	14,35,37,5	7,27,29,33,11,13,31	26	"	750	"	"
P7	DB1 "	"	14,35,37,7	5,27,29,33,11,13,31	6	"	"	"	"
P8	DB2 "	"	14,35,37,27	5,7,29,33,11,13,31	28	"	"	"	"
P9	DB3 "	"	14,35,37,29	5,7,27,33,11,13,31	30	"	"	"	"
P10	DB4 "	"	14,35,37,33	5,7,27,29,11,13,31	34	"	"	"	"
P11	DB5 "	"	14,35,37,11	5,7,27,29,33,13,31	8	"	"	"	"
P12	DB6 "	"	14,35,37,13	5,7,27,29,33,11,31	12	"	"	"	"
P13	DB7 "	"	14,35,37,31	5,7,27,29,33,11,13	32	"	"	"	"

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TABLE XVI. Crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P14	D0 Crosstalk, low level	14,35	36,37,26	12,32,6,28,30,34,8	5	0	650	mV	20,22
P15	D1 "	"	36,37,6	12,32,26,28,30,34,8	7	"	"	"	"
P16	D2 "	"	36,37,28	12,32,26,6,30,34,8	27	"	"	"	"
P17	D3 "	"	36,37,30	12,32,26,6,28,34,8	29	"	"	"	"
P18	D4 "	"	36,37,34	12,32,26,6,28,30,8	33	"	"	"	"
P19	D5 "	"	36,37,8	12,32,26,6,28,30,34	11	"	"	"	"
P20	D6 "	"	36,37,12	8,32,26,6,28,30,34	13	"	"	"	"
P21	D7 "	"	36,37,32	8,12,26,6,28,30,34	31	"	"	"	"

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TABLE XVII. Crosstalk (high-level) test.

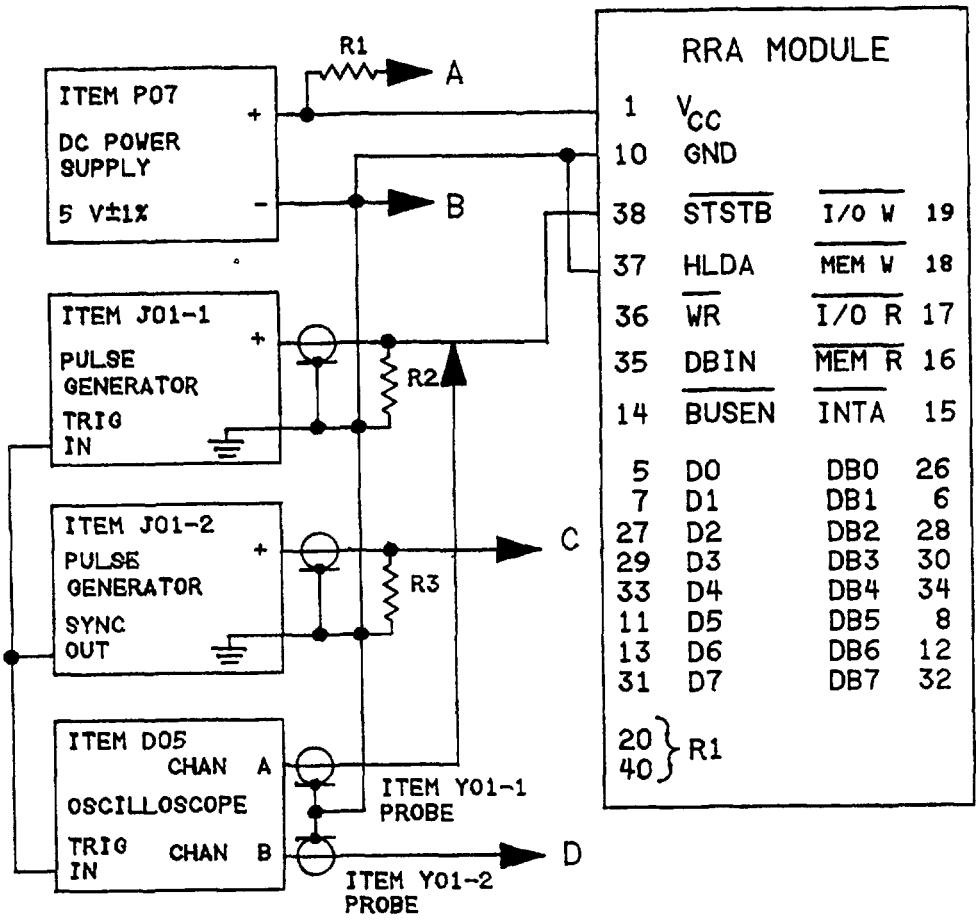
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	14,35,36,37	33,11,13,31,5,7,27,29	Open	---	---	---	20,21
Q1	$\overline{I/OR}$ Crosstalk, high level	"	"	"	17	0	400	mV	20,22
Q2	$\overline{I/OW}$ "	"	"	"	19	"	"	"	"
Q3	$\overline{MEMR}$ "	"	"	"	16	"	"	"	"
Q4	$\overline{INTA}$ "	"	"	"	15	"	"	"	"
---	Setup	31	27,29,13,7,14,35,37,5	Open	Open	---	---	---	20,21
Q5	$\overline{MEMW}$ Crosstalk, high level	Open	14,35,37	33,11,13,31,5,7,27,29	18	0	400	mV	20,23
Q6	DB0 "	5	"	7,27,29,33,11,13,31	26	"	"	"	"
Q7	DB1 "	7	"	5,27,29,33,11,13,31	6	"	"	"	"
Q8	DB2 "	27	"	5,7,29,33,11,13,31	28	"	"	"	"
Q9	DB3 "	29	"	5,7,27,33,11,13,31	30	"	"	"	"
Q10	DB4 "	33	"	5,7,27,29,11,13,31	34	"	"	"	"
Q11	DB5 "	11	"	5,7,27,29,33,13,31	8	"	"	"	"
Q12	DB6 "	13	"	5,7,27,29,33,11,31	12	"	"	"	"
Q13	DB7 "	31	"	5,7,27,29,33,11,13	32	"	"	"	"
Q14	D0 "	14,35,26	36,37	12,32,6,28,30,34,8	5	"	"	"	"
Q15	D1 "	14,35,6	"	12,32,26,28,30,34,8	7	"	"	"	"
Q16	D2 "	14,35,28	"	12,32,26,6,30,34,8	27	"	"	"	"
Q17	D3 "	14,35,30	"	12,32,26,6,28,34,8	29	"	"	"	"
Q18	D4 "	14,35,34	"	12,32,26,6,28,30,8	33	"	"	"	"

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TABLE XVII. Crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q19	D5	Crosstalk, high level	14,35,8	36,37	12,32,26,6,28,30,34	11	0	400	mV	20,22
Q20	D6	"	14,35,12	"	8,32,26,6,28,30,34	13	"	"	"	"
Q21	D7	"	14,35,32	"	8,12,26,6,28,30,34	31	"	"	"	"

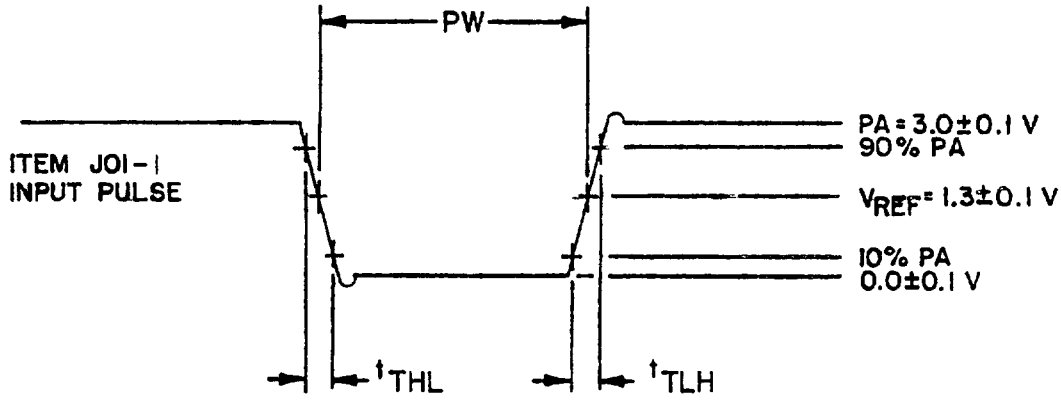
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NOTES
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. For the setup conditions, place item J01-1 in single pulse mode and pulse the module one time with a pulse as shown on figure 22.
4. Load all outputs with a load as shown on figure 14.
5. Item J01-1 is used for setup conditions only.

FIGURE 20. Crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

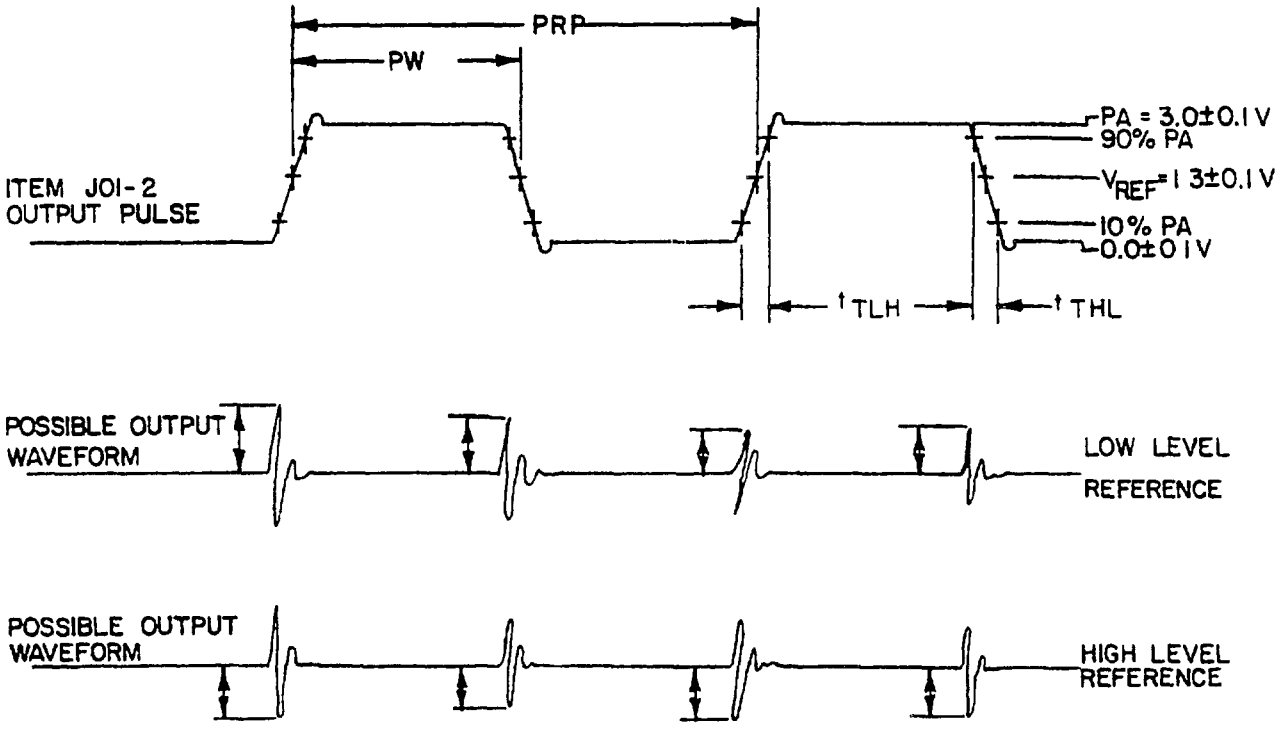
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NOTES.
1. Input pulse parameters
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns.
   Single pulse mode.
2 Using item JOI-1, apply one pulse after establishing the setup conditions in tables XVI and XVII.

FIGURE 21 Crosstalk (low-level and high-level) test circuit waveform for tables XVI and XVII.

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NOTES
1. Input pulse parameter.
   PW = 200 ±20 ns
   PRP = 400 ±40 ns
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns
2. Measure the most positive peak from the low level reference for the low level crosstalk test and the most negative peak from the high level reference for the high level crosstalk test.

FIGURE 22. Crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	23,24

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S	Power supply filter capacitance	1	Open	1.6	3.0	µF	25

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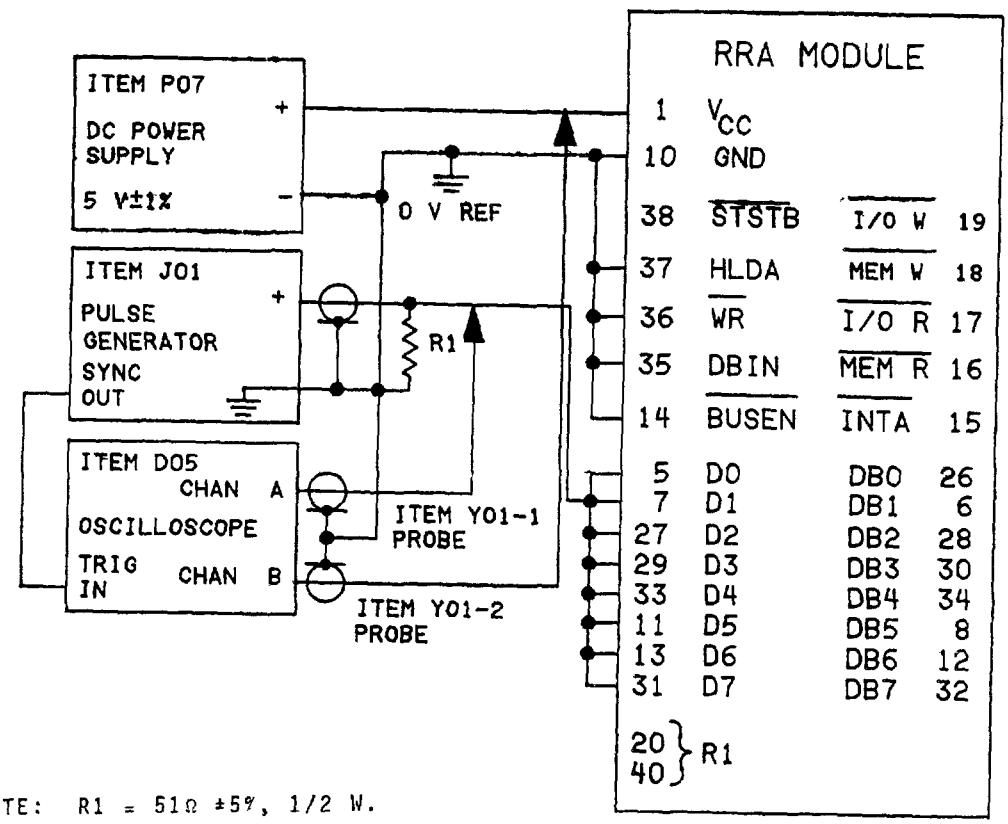
TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	$\overline{\text{STSTB}}$ Input capacitance	Open	38	2	20	pF	25
T2	HLDA "	"	37	"	"	"	"
T3	$\overline{\text{WR}}$ "	"	36	"	"	"	"
T4	DBIN "	"	35	"	"	"	"
T5	$\overline{\text{BUSEN}}$ "	"	14	"	"	"	"

TABLE XXI. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	$\overline{\text{I/OW}}$ Output capacitance	Open	19	2	23	pF	25
U2	$\overline{\text{MEMW}}$ "	"	18	"	"	"	"
U3	$\overline{\text{I/OR}}$ "	"	17	"	"	"	"
U4	$\overline{\text{MEMR}}$ "	"	16	"	"	"	"
U5	$\overline{\text{INTA}}$ "	"	15	"	"	"	"

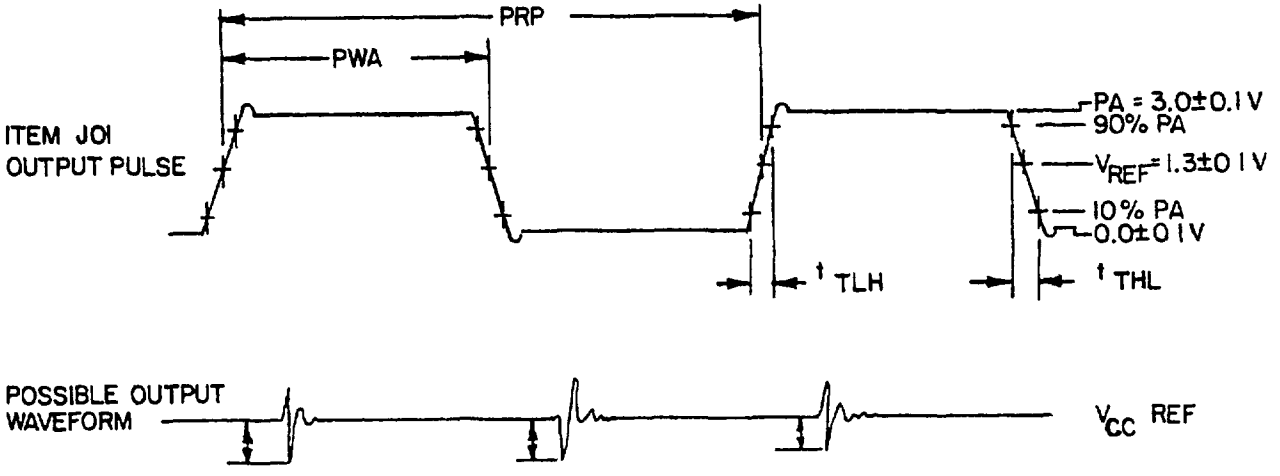
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 23 Power supply transient amplitude test circuit for table XVIII.

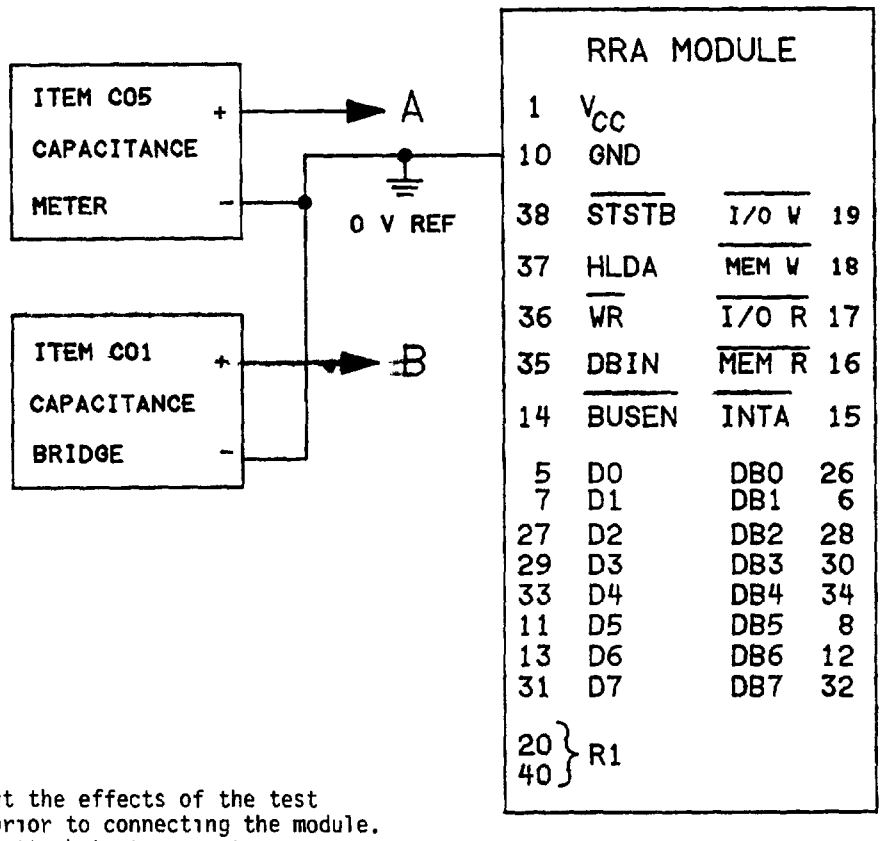
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NOTES:
1. Input pulse parameters.
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. Measure the most negative peak referenced from V_{CC}.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3 Item C01 setup
Frequency = 1 kHz,
V_{pk} = 0 3 V maximum.

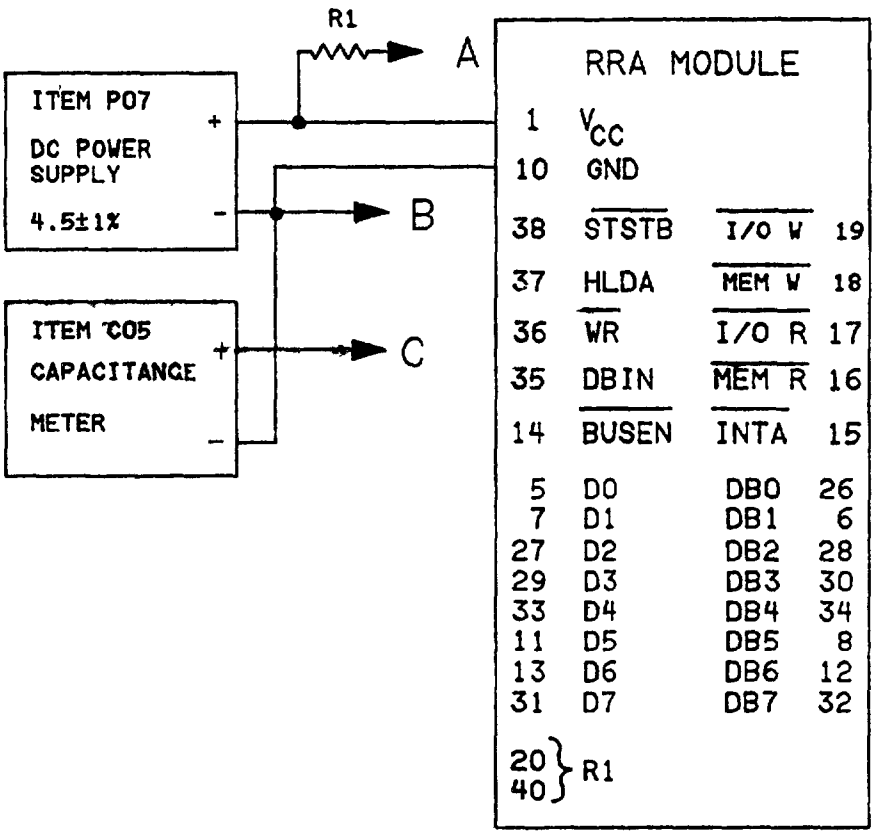
FIGURE 25 Power supply filter capacitance, input capacitance, and output capacitance test circuit for tables XIX - XXI.

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TABLE XXII. Input/output capacitance test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	D0 Input/output capacitance	Open	14,35,37	5	3	35	pF	26
V2	D1 "	"	"	7	"	"	"	"
V3	D2 "	"	"	27	"	"	"	"
V4	D3 "	"	"	29	"	"	"	"
V5	D4 "	"	"	33	"	"	"	"
V6	D5 "	"	"	11	"	"	"	"
V7	D6 "	"	"	13	"	"	"	"
V8	D7 "	"	"	31	"	"	"	"
V9	DB0 "	14,35	37	26	"	"	"	"
V10	DB1 "	"	"	6	"	"	"	"
V11	DB2 "	"	"	28	"	"	"	"
V12	DB3 "	"	"	30	"	"	"	"
V13	DB4 "	"	"	34	"	"	"	"
V14	DB5 "	"	"	8	"	"	"	"
V15	DB6 "	"	"	12	"	"	"	"
V16	DB7 "	"	"	32	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. R1 = 1 kΩ ±1%, 1/8 W.

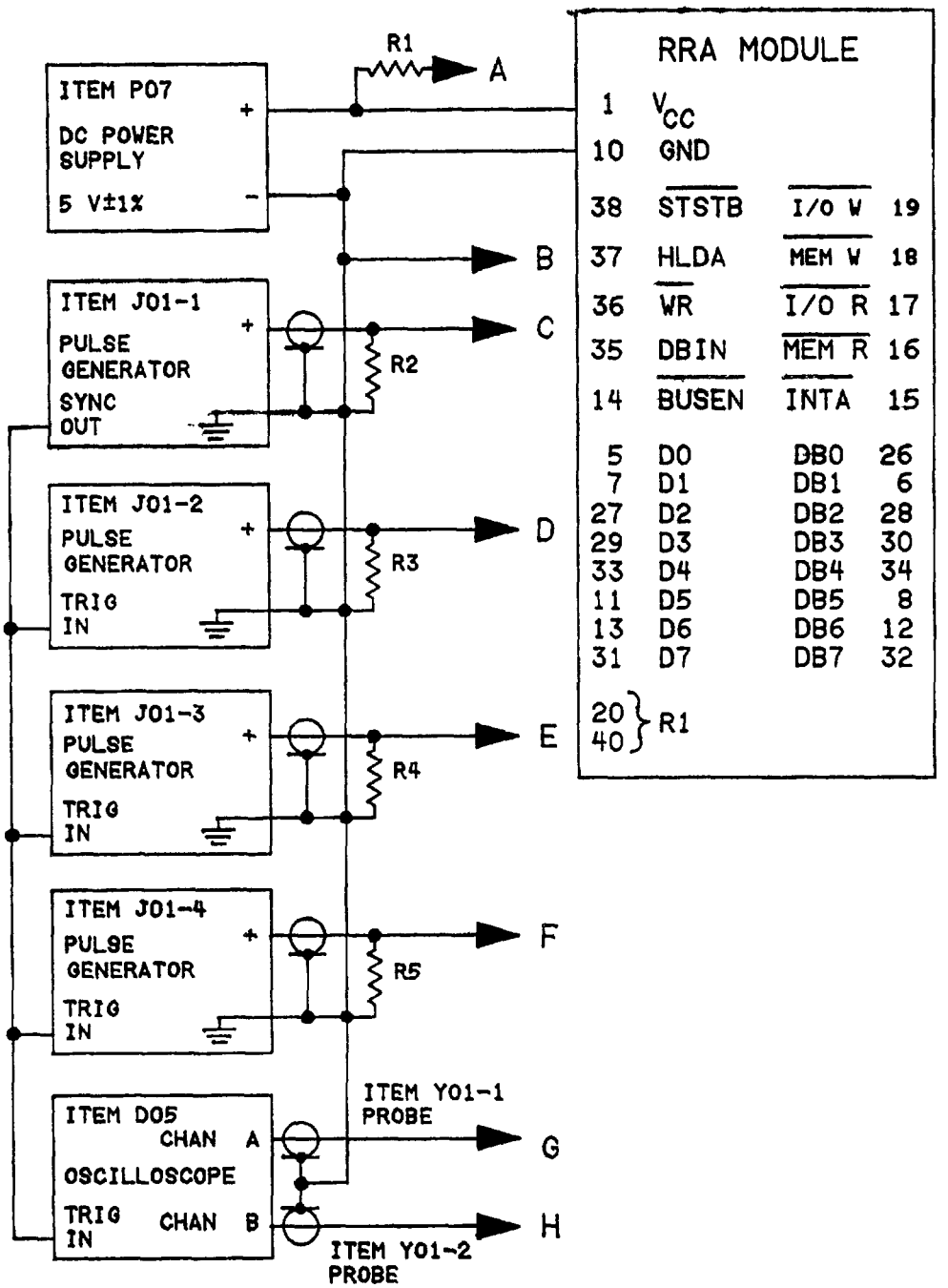
FIGURE 26. Input/output capacitance test circuit for table XXII.

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TABLE XXIII. Setup and hold time test.

Test condition	Parameter	Connect								Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
W1	$\overline{\text{STSTB}}$ to $\overline{\text{INTA}}$ Setup and hold time	36,37	14	38	7,13	5,27,11	35	38	15	20	80	ns	27,28
W2	$\overline{\text{STSTB}}$ to $\overline{\text{MEMR}}$ "	"	"	"	5,7,27, 29,11	13,31	"	"	16	"	"	"	"
W3	$\overline{\text{STSTB}}$ to $\overline{\text{I/OR}}$ "	"	"	"	27	13	"	"	17	"	"	"	"
W4	DBIN to $\overline{\text{INTA}}$ "	Open	14,37, 27	"	Open	5,11	"	35	15	4	45	"	27,29
W5	DBIN to $\overline{\text{MEMR}}$ "	"	"	"	5,29	11,31	"	"	16	"	"	"	"
W6	DBIN to $\overline{\text{I/OR}}$ "	"	"	"	Open	11,13	"	"	17	"	"	"	"
W7	HLDA to $\overline{\text{INTA}}$ "	35	14,27	"	Open	5,11	37	37	15	4	40	"	27,30
W8	HLDA to $\overline{\text{MEMR}}$ "	"	"	"	5,29	11,31	"	"	16	"	"	"	"
W9	HLDA to $\overline{\text{I/OR}}$ "	"	"	"	Open	13	"	"	17	"	"	"	"
W10	$\overline{\text{WR}}$ to $\overline{\text{MEMW}}$ "	Open	14,27, 7	"	33	Open	36	36	18	7	70	"	27,31
W11	$\overline{\text{WR}}$ to $\overline{\text{MEMW}}$ "	"	14,27, 7	"	33	Open	"	"	18	"	"	"	"
W12	$\overline{\text{WR}}$ to $\overline{\text{I/OW}}$ "	"	14,27	"	Open	33	"	"	19	"	"	"	"
W13	$\overline{\text{WR}}$ to $\overline{\text{I/OW}}$ "	"	14,27	"	Open	33	"	"	19	"	"	"	"

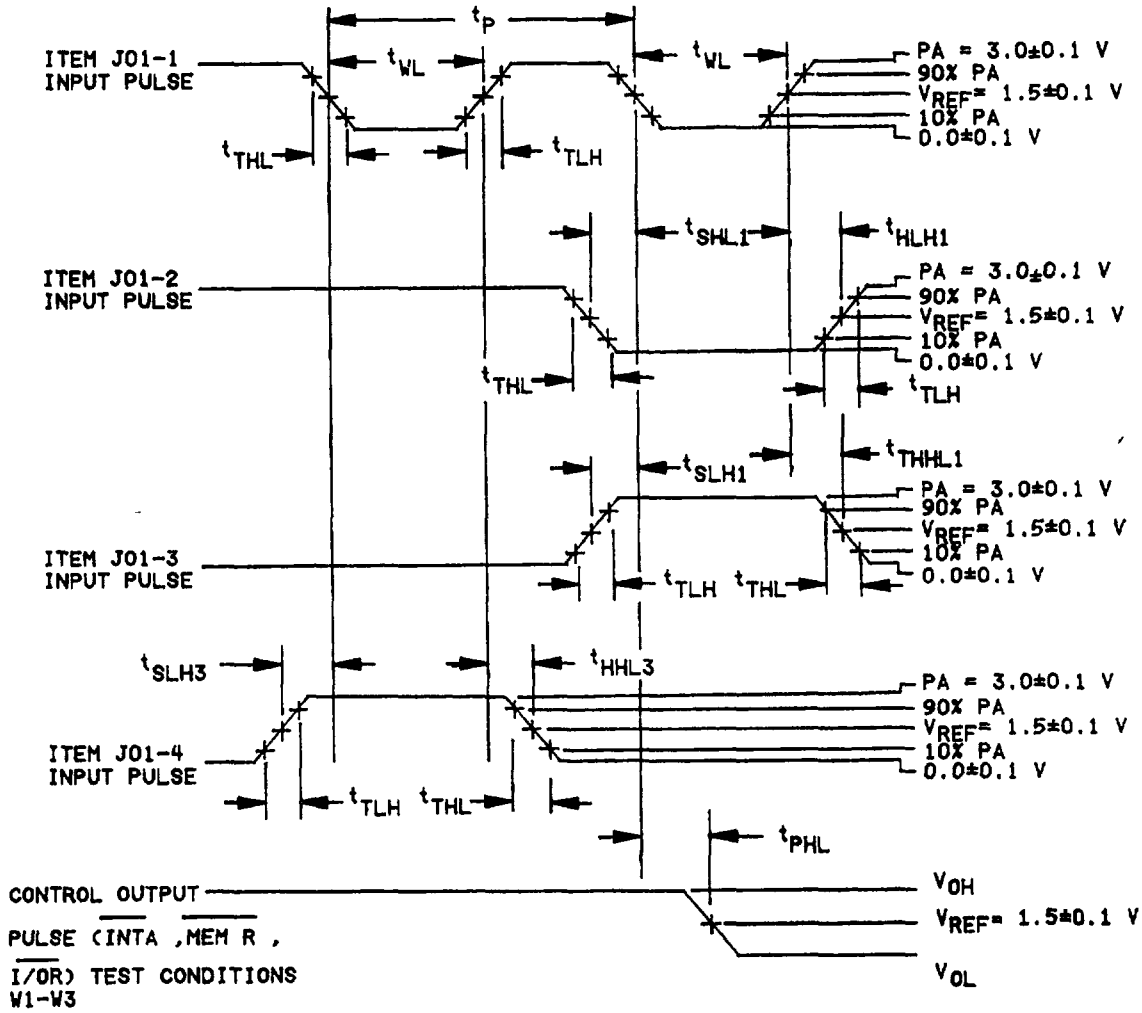
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NOTES
1 R1 = 1 kΩ ±1%, 1/8 W
2 R2 - R5 = 51Ω ±5%, 1/2 W.
3 Load all outputs with a load as shown on figure 14.

FIGURE 27. Setup and hold time test circuit for table XXIII

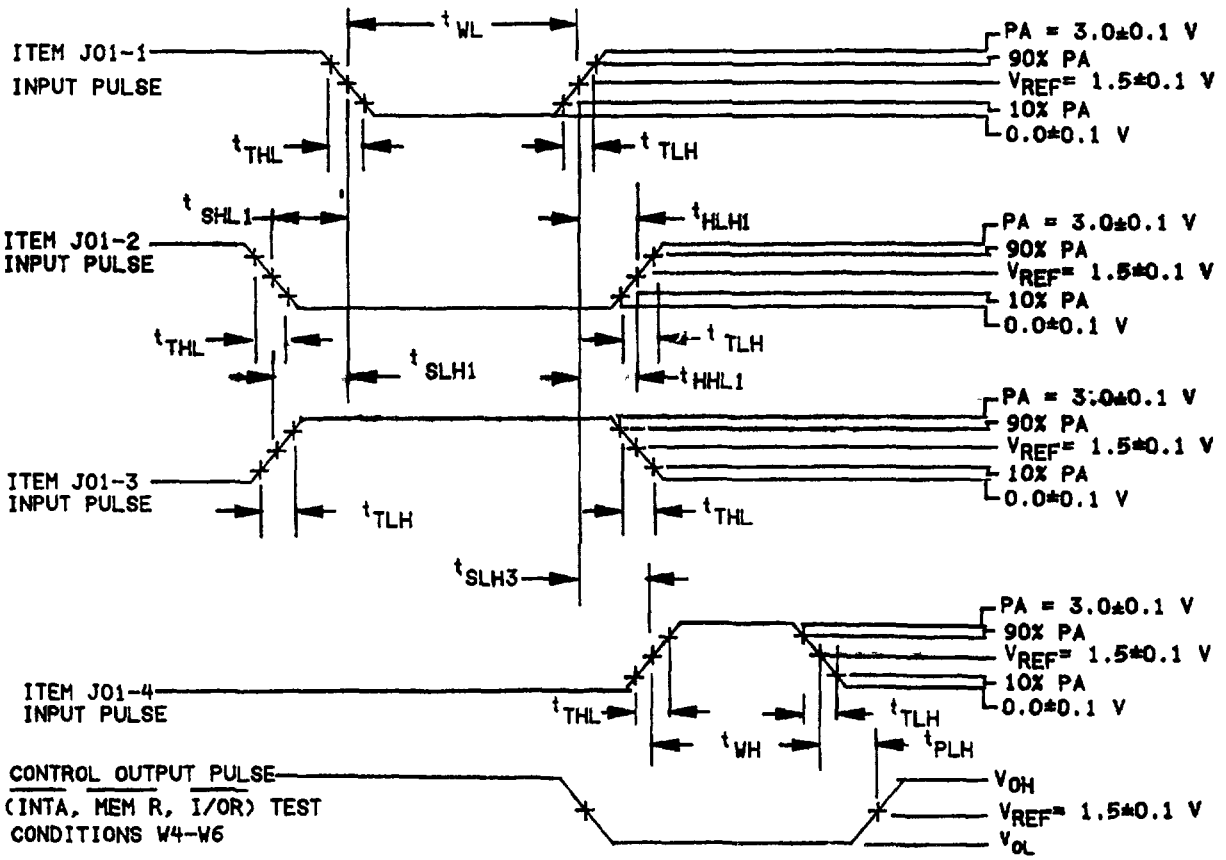
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   t_{WL} = 25 +2, -0 ns
   t_P = 100 ±10 ns
   $t_{SLH1} = t_{SHL1}$ = 12 +2, -0 ns
   $t_{HLH1} = t_{HHL1}$ = 5 +2, -0 ns.
   t_{SLH3} = 20 ±2 ns.
   t_{HHL3} = 20 ±2 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making the specified measurements, verify that the output high level is $\geq$2 4 V and the output low level is $\leq$0 5 V

FIGURE 28. Setup and hold time test circuit waveforms for table XXIII.

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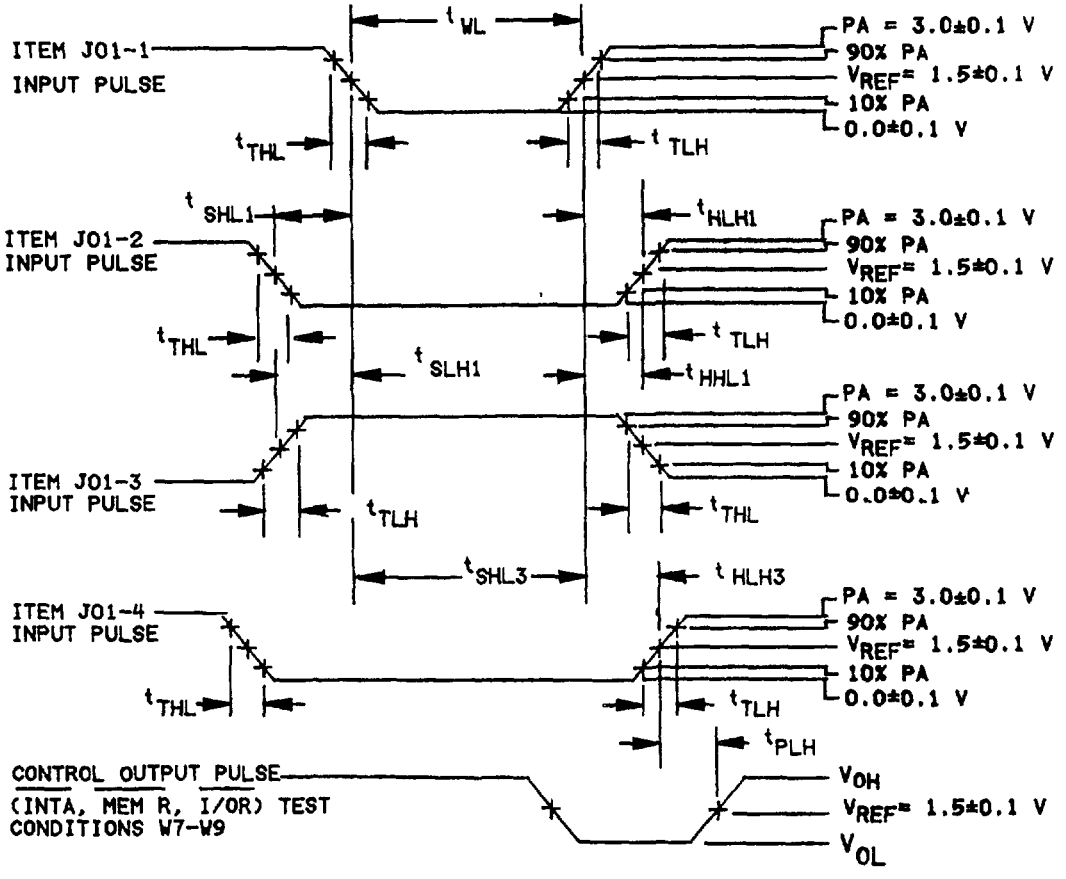


NOTES

1. Input pulse parameters:
   t_{WL} = 25 +2, -0 ns
   PRP (item J01-1) = 100 ±10 ns.
   $t_{SHL1} = t_{SLH1}$ = 12 +2, -0 ns.
   $t_{HLH1} = t_{HHL1}$ = 5 +2, -0 ns
   t_{SLH3} = 20 ±2 ns.
   t_{WH} = 30 ±5 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the specified measurements, verify that the output high level is ≥2.4 V and the output low level is ≤0.5 V.

FIGURE 29. Setup and hold time test circuit waveforms for table XXIII.

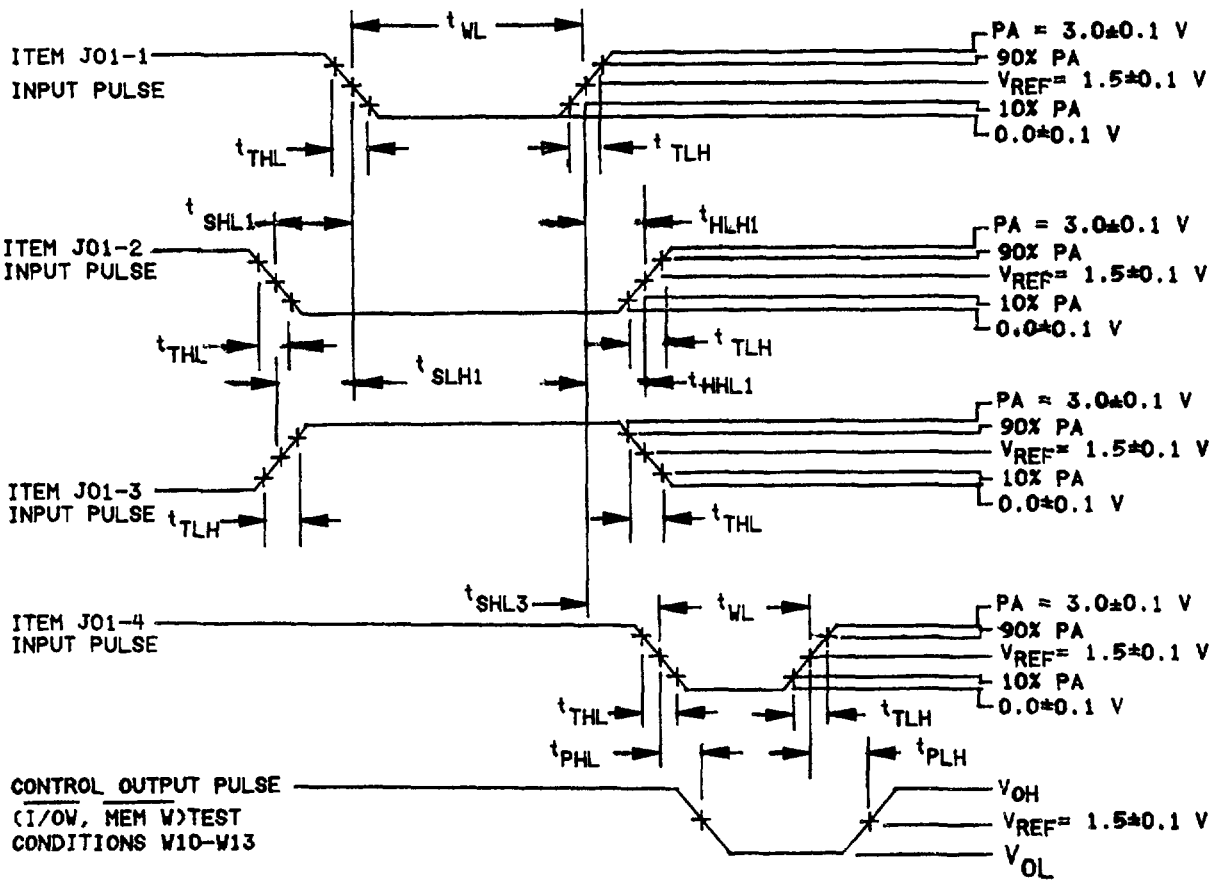
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   t_{WL} = 25 +2, -0 ns.
   PRP (item J01-1) = 150 ±15 ns.
   $t_{SHL1} = t_{SLH1}$ = 12 +2, -0 ns.
   $t_{HLH1} = t_{HHL1}$ = 5 +2, -0 ns.
   t_{SHL3} = 20 ±2 ns.
   t_{HLH3} = 20 ±2 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the specified measurements, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.5 V.

FIGURE 30. Setup and hold time test circuit waveforms for table XXIII.

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NOTES

1. Input pulse parameters
   t_{WL} = 25 +2, -0 ns.
   PRP (item J01-1) = 100 ±10 ns.
   $t_{SHL1} = t_{SLH1}$ = 12 +2, -0 ns
   $t_{HLH1} = t_{HHL1}$ = 5 +2, -0 ns.
   t_{SHL3} = 25 ±5 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making the specified measurements, verify that the output high level is ≥2 4 V and the output low level is ≤0.5 V.

FIGURE 31. Setup and hold time test circuit waveforms for table XXIII

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TABLE XXIV. Setup and hold time test.

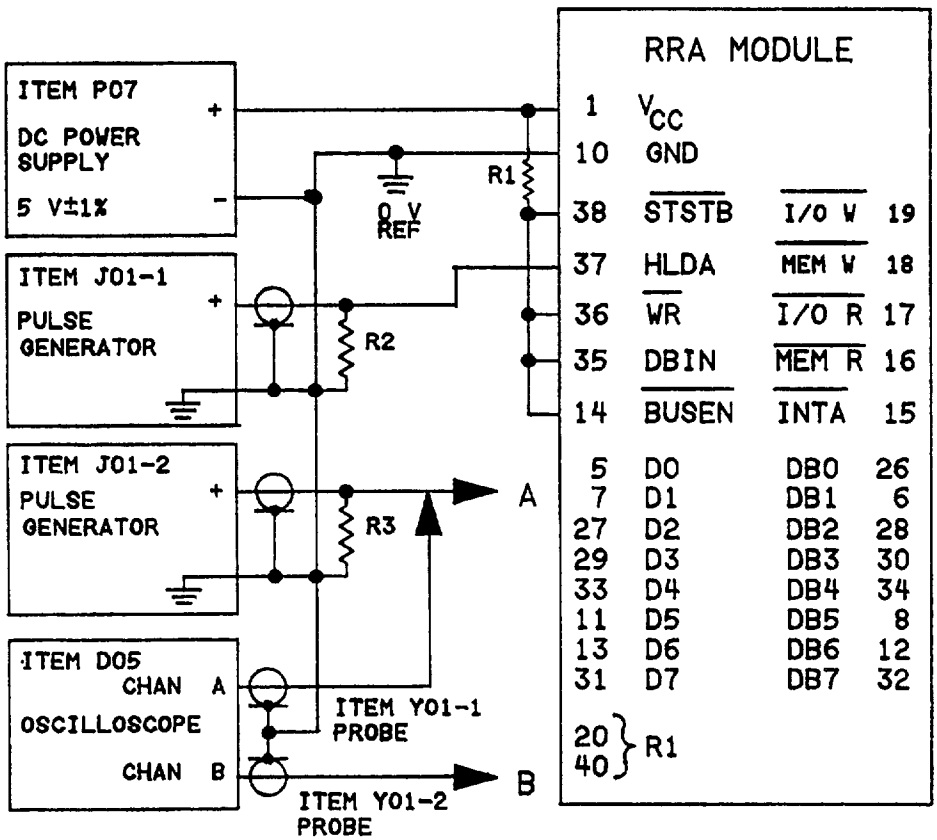
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	DB0-HLDA Setup and hold time (t_{SLH})	26	5	10	100	ns	32,33
X2	DB1-HLDA "	6	7	"	"	"	"
X3	DB2-HLDA "	28	27	"	"	"	"
X4	DB3-HLDA "	30	29	"	"	"	"
X5	DB4-HLDA "	34	33	"	"	"	"
X6	DB5-HLDA "	8	11	"	"	"	"
X7	DB6-HLDA "	12	13	"	"	"	"
X8	DB7-HLDA "	32	31	"	"	"	"
X9	DB0-HLDA Setup and hold time (t_{HHL})	26	5	"	"	"	"
X10	DB1-HLDA "	6	7	"	"	"	"
X11	DB2-HLDA "	28	27	"	"	"	"
X12	DB3-HLDA "	30	29	"	"	"	"
X13	DB4-HLDA "	34	33	"	"	"	"
X14	DB5-HLDA "	8	11	"	"	"	"
X15	DB6-HLDA "	12	13	"	"	"	"
X16	DB7-HLDA "	32	31	"	"	"	"
X17	DB0-HLDA Setup and hold time (t_{SHL})	26	5	"	"	"	32,34
X18	DB1-HLDA "	6	7	"	"	"	"
X19	DB2-HLDA "	28	27	"	"	"	"
X20	DB3-HLDA "	30	29	"	"	"	"
X21	DB4-HLDA "	34	33	"	"	"	"

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TABLE XXIV. Setup and hold time test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X22	DB5-HLDA Setup and hold time (t_{SHL})	8	11	10	100	ns	32,34
X23	DB6-HLDA "	12	13	"	"	"	"
X24	DB7-HLDA "	32	31	"	"	"	"
X25	DB0-HLDA Setup and hold time (t_{HLH})	26	5	"	"	"	"
X26	DB1-HLDA "	6	7	"	"	"	"
X27	DB2-HLDA "	28	27	"	"	"	"
X28	DB3-HLDA "	30	29	"	"	"	"
X29	DB4-HLDA "	34	33	"	"	"	"
X30	DB5-HLDA "	8	11	"	"	"	"
X31	DB6-HLDA "	12	13	"	"	"	"
X32	DB7-HLDA "	32	31	"	"	"	"

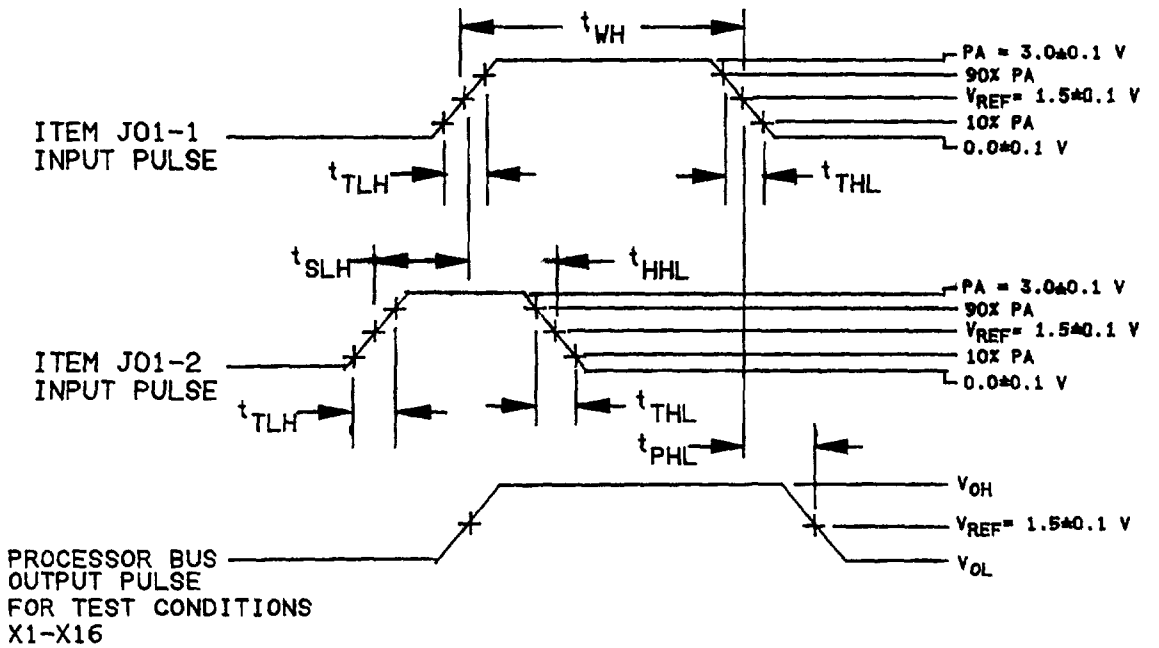
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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W
3. Load all outputs with a load as shown on figure 14

FIGURE 32. Setup and hold time test circuit for table XXIV.

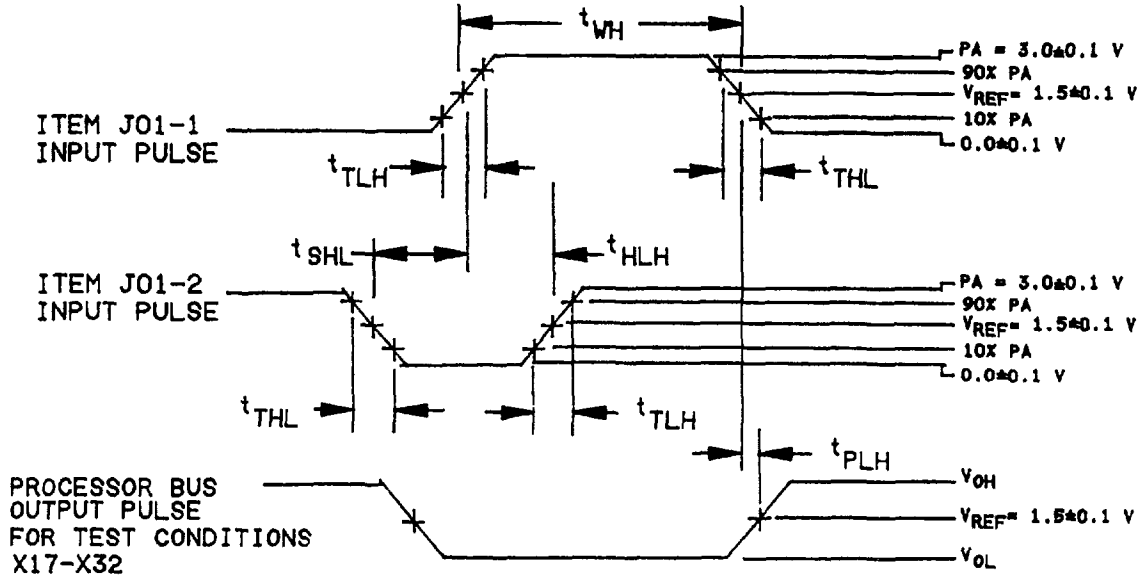
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NOTES:
1. Input pulse parameters:
   t_{WH} = 200 ±20 ns.
   PRP (item JO1-1) = 400 ±40 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_{SLH} = 10 +2, -0 ns for test conditions X1-X8.
   = 20 ±2 ns for test conditions X9-X16.
   t_{HHL} = 40 ±4 ns for test conditions X1-X8.
   = 20 ±2, -0 ns for test conditions X9-X16.
2. Before making the specified measurements, verify that the output high level is $\geq$ 3.3 V and the output low level is $\leq$ 0.5 V.

FIGURE 33. Setup and hold time test circuit waveforms for table XXIV.

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NOTES:
1. Input pulse parameters:
   t_{WH} = 200 ±20 ns.
   PRP (item JO1-1) = 400 ±40 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_{SHL} = 10 +2, -0 ns for test conditions X17-X24.
   = 20 ±2 ns for test conditions X25-X32.
   t_{HLH} = 40 ±4 ns for test conditions X17-X24.
   = 20 ±2, -0 ns for test conditions X25-X32.
2. Before making the specified measurements, verify that the output high level is $\geq$ 3.3 V and the output low level is $\leq$ 0.5 V.

FIGURE 34. Setup and hold time test circuit waveforms for table XXIV.

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